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Using machine learning to predict student science achievement based on science curriculum type in TIMSS 2019

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ABSTRACT

Most educational systems use either an integrated or a separated science curriculum. However, it is unclear which of these science curricula benefits students more and existing research provides insufficient information about the implementation details of the curriculum employed. Therefore, this study compares the effects of two science curricula on students' science literacy, drawing on socio-ecological theory and employing educational data mining techniques. Results from Grade 8 Science students in 44 countries sampled in the Trends in International Mathematics and Science Study (TIMSS) 2019 showed that (1) the integrated curricula benefitted students marginally more than the separated curricula; (2) curriculum type was not essential in directly predicting students' academic performance; and (3) random forest outperformed linear regression, lasso regression, decision trees, and neural networks in predicting student science achievement. This study advances our understanding of the predictors of student science performance, demonstrates that machine learning techniques can be applied successfully to examine curriculum effects, and provides directions for implementing integrated science curricula.

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Science curriculum; TIMSS; educational data mining; machine learning; propensity score matching

Introduction

The advancement of technology in the twenty-first century has been accompanied by an avalanche of data, information, and knowledge, which has led to the rapid development of all aspects of society and increased global communication and cooperation. However, this has also brought about many contemporary unique issues, such as climate change and rapidly spreading diseases. **These modern distinct challenges call for citizens to 'learn integrated and usable knowledge, rather than the sets of compartmentalized and decontextualized facts'** (Sawyer, 2022, p. 2). Moreover, schools should be ready to prepare students for this integrated thinking to help them develop competence when entering the workforce. Correspondingly, many educational systems have already implemented integrated science curricula rather than traditionally separated science curricula (Haggis & Adey, 1979). The former is believed to have the capacity to foster

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students' holistic thinking to address contemporary environmental and societal issues (Lee & Wan, 2022; Nagle, 2013; Venville et al., 2012), whereas the latter places less emphasis on cultivating this kind of ability in students (Wei, 2018).

Moreover, students learn more deeply when provided with authentic contexts (Sawyer, 2022). An integrated curriculum is more capable of providing authentic practices than a separated science curriculum (Lee & Wan, 2022). For example, an integrated curriculum for science regarding climate change (Nagle, 2013, p. 1) blends 'concepts in the earth sciences (atmosphere, weather, and climate), physics (heat and temperature), and life sciences (effects of climate change on ecosystems).' In contrast, a separated science curriculum often distributes the relevant concepts across several subjects. For instance, the educational system in Finland deploys climate change-related concepts in Chemistry (e.g. sustainable consumption and products lifecycle), Physics Biology (e.g. the habitats of animals and observing changes in the environment), and Geography (e.g. lifecycle assessment and limited natural resources) (Dawson et al., 2022).

In this study, integrated science curricula enable students to learn interdisciplinary topics within a single class. For instance, students can learn concepts of earth sciences, physics, and life sciences in the same unit of an integrated curriculum that targets climate change. Conversely, the separated science curricula enable students to learn topics as part of individual courses. For example, students can learn Physics topics from a dedicated Physics course.

However, there is no consensus regarding which science curriculum benefits students more. Several researchers advocated for the integrated science curriculum (e.g. Beichner et al., 1999; Czerniak & Johnson, 2015; Thang & Koh, 2017), whereas others criticised it (e.g. Ogunkola & Samuel, 2011; Sun et al., 2014). This leads to the following research question: **Which type of science curriculum helps students achieve more in science?** Answering this question may help deploy a science curriculum in K-12 education that can prepare students for a competitive environment in a society that relies increasingly on scientific knowledge. This study used propensity score matching and machine learning methods to investigate the effect of two science curricula in addressing the research question.

Separated curriculum

Separated curriculum instruction has been the mainstream from the start of science education (Brown, 1977; Wei, 2009). Universities have had well-established programmes for training individual science discipline teachers. Secondary education has developed separated science curriculum frameworks to prepare students for high-stakes examinations (e.g. the university entrance exam). In most nations, students must follow four major science curricula: Biology, Physics, Chemistry, and Earth Science.

However, such a segmented learning experience is not well positioned to help students form a holistic understanding of contemporary 'scientific perplexities,' such as environmental sustainability (Venville et al., 2012). The outbreak of Coronavirus constituted an example that stressed the importance of a holistic understanding of science. In particular, Linn et al. (2016) asserted that students cannot coherently understand and internalise more than 60 distinct science concepts or topics in a year. Due to the inability to link segmented curricula with job expectations, many students regard science as useless

(Wei, 2009). Meanwhile, schools or regions often choose a few science disciplines as the essential teaching and learning target due to limited teaching time but neglect others in many regions. For example, Wan and Lee (2021) concluded that Korea and Taiwan (China) both emphasise Physics topics more in science textbooks. Au (2007) also argued that teachers often prioritise concepts that are covered in high-stakes tests when delivering instructional materials.

Several instructional frameworks were proposed, including knowledge integration (Linn et al., 2016), aiming to make science more practically meaningful for students. It is believed that an integrated curriculum allows students to build a holistic thinking pattern by considering real-world contexts (Lee & Wan, 2022; Venville et al., 2012).

Integrated curriculum

TIMSS 2019 reported the science curriculum type adopted by each participant country or region. Combining these statistics, more than 60% of countries or regions implemented integrated science curricula, while others applied separated science curricula.

Linn et al. (2016) classified a century of science education history in America into four stages: separation (1916–1960), interaction (1960–1980), partnership (1980–1995), and integration (1995–2016). Such a development pattern reflects the common trends in the history of science education. Many official standards have been inspired by the promised benefits of integrated science. For example, the United States published the 2011 Framework for K–12 Science Education, which includes the ‘cross-cutting concepts’ in science curriculum standards (National Research Council, 2012). Therefore, combining different subjects into one curriculum is a common solution to address several drawbacks of separated curricula. Such integration could remove content repetition between subjects and save teaching time (Brown, 1977), while increasing the number of covered topics in disciplines that have long suffered from neglect.

However, an integrated curriculum requires much more development than a separated curriculum to benefit students. The need for such additional effort arises from the challenging fulfilment of the prerequisites of a successfully integrated curriculum in practice. First, compiling a coherent textbook for the integrated curriculum can be laborious (Scotter et al., 2000; Wan & Lee, 2021; Wang, 2018). Wan and Lee (2021) argued that the design of the textbook, curriculum, and syllabus should be consistent with the inner logic of that discipline. Nevertheless, an examination of integrated science textbooks in Korea and Taiwan (China) suggested that coherence is only partially guaranteed (Lee & Wan, 2022). Linn et al. (2016) agreed with Bruner’s (1960) standpoint that science topics need to be sequenced by their abstraction level. Therefore, Linn et al. (2016) also argued that selecting and sequencing topics for science textbooks are the most prominent challenges in science instruction. Moreover, Pearson et al. (2010) argued that the poor quality of science textbooks hinders the proliferation of integrated science instruction.

Second, a qualified teacher is essential for the success of integrated science (e.g. Lee & Wan, 2022; Wei, 2018). Teachers with a background in only one discipline find it challenging to teach four subjects (Wei, 2009). One of the solutions is to assign different instructors to teach different parts of an integrated curriculum but leave the scientific design of the structure and sequence of the integrated curriculum content ignored in practice (Wei, 2009). In turn, this approach may hinder students’ ability to link topics

from different subjects. Training instructors to be competent in teaching integrated curricula with multiple science discipline backgrounds is also complex and time-consuming (Harrell, 2010). Moreover, teachers tend to prioritise knowledge topics that carry more weight in high-stakes examinations when presenting integrated science materials (Nagle, 2013).

Third, the lagging development of measurement instruments that do not comply with the new form of the curriculum also obstructs the implementation of the integrated curriculum. Moreover, accessing the necessary resources for the integrated science curriculum in economically disadvantaged regions is relatively challenging.

In summary, successfully implementing an integrated science curriculum requires the ‘transformation of the curriculum, instruction, assessments, and teacher professional development’ (Nagle, 2013). Although integrated curricula are gradually being adopted around the globe (Haatainen et al., 2021), there has yet to be an agreement on which science curriculum benefits students more. Some studies indicated that integrated instruction cultivates students’ confidence and satisfaction with science (Beichner et al., 1999; Hurley, 2001). In contrast, another research branch asserts that integrated science is not as helpful as it promised (Meichtry, 1993; Sun et al., 2014; Venville et al., 2012).

Research gap

There is considerable discrepancy in research on the effects of the two curricula on student achievement. Several studies advocate for the benefits of an integrated science curriculum for students’ science achievement, attitudes toward science, and scientific skills (e.g. Alake & Ogunseemi, 2013; Beichner et al., 1999; Czerniak & Johnson, 2015; Thang & Koh, 2017). Other studies found that integrated science did not contribute to student learning and attitudes and that integrated science teachers are not always qualified to teach it (e.g. Harrell, 2010; Ogunkola & Samuel, 2011; Sun et al., 2014).

However, these studies report insufficient information about the implementation details of the selected integrated science curriculum, such as the teacher’s academic background and the instructional time employed. Consequently, comparing the implementation details between different studies that reach the opposite conclusion is complicated due to inadequate information provided. Furthermore, the comparison of the two curricula is biased without controlling the instructional detail-related variable. In addition, existing studies are often restricted to small sample sizes, making the conclusion less general. Although these results might be suited for their sampled areas, the effect of curriculum needs to be explored more generally across multiple geographical regions. Therefore, it is necessary to investigate curriculum effects on student learning outcomes by using large-scale assessment data and controlling instructional detail-related variables. In contrast to locally collected data, TIMSS provides curriculum data and, thus, a chance to examine the curriculum effects beyond the district restrictions.

The current study

Educational practices have far-reaching consequences for many societal aspects. The curriculum setting is closely related to the standard curriculum alignment, science teacher

demand, textbook revision, measurement instrument creation, student academic performance, and many other critical educational components. Moreover, these factors that partially depend on the curriculum settings are also weaving together, making the problem more complex. In 1985, Hodson (1985) pointed out that the choice between integrated and separated science curricula should be made at the outset. However, up to the present, the research findings could not provide definite advantages and disadvantages for all the stakeholders in this interwoven chain.

Given that educational researchers must provide unbiased support for their curriculum choice, this project aims to explore this question based on the educational data mining paradigm, which could provide novel insights into this long-standing debate. We pose two research questions:

- (1) How do the two curricula effects on student achievement compare in TIMSS 2019?
- (2) Does science curriculum type predict student achievement in science?

The rest of the manuscript is organised as follows. It starts with a literature review summarising the findings and methods of related studies to provide the background for this research, followed by the theoretical framework that supports this study. Then, it presents the methods and results, followed by a discussion, implications, and conclusions.

Literature review

Integrated and separated science curricula

Modern science education traditionally classifies nature science into Biology, Chemistry, Earth Science, and Physics. Many countries and regions have implemented this curriculum design, known as the separated science curriculum, for a long time.

The twentieth century saw the emergence of many integrated science curricula. Many conferences and symposiums surrounding the evolution of integrated science curricula were held, such as ‘Coordination of the Teaching of Mathematics and Physics’ in 1967 and conferences in 1968, 1973, 1975, and 1978, which were held by the United Nations Educational Scientific and Cultural Organization (UNESCO) and other institutions (Frey, 1989). Following these conferences, UNESCO published a series of reports about integrated science curricula named ‘New Trends in Integrated Science Teaching’ (Chisman, 1990).

This study analyzes the TIMSS data set. Given that the participants of TIMSS 2019 include many countries and regions around the globe, selecting the most general definition of integrated science would be appropriate to all places adopting an integrated science curriculum: ‘An integrated science curriculum contains three or more science discipline topics from Biology, Earth science, Physics, and Chemistry’ (Lee & Wan, 2022).

Essential factors for successful integrated science implementation

As Rutherford (2000) indicated, an integrated science curriculum could only be helpful after substantial development. Therefore, it is necessary and critical to evaluate whether its goal can be achieved through intentional implementation (Lee & Wan, 2022). When

the necessary prerequisites for an effective curriculum are not met, even the most promising design will not suffice. Several studies pointed out that the essential factors for the success of integrated science include a qualified teacher (e.g. Lee & Wan, 2022; Wei, 2018) and a coherent textbook (e.g. Scotter et al., 2000; Wan & Lee, 2021). Venville et al. (2012) argued that administrative policy, testing constraints, and school traditions could also impact the success or failure of integrated science. For example, the assessment of an integrated curriculum tends to overlook its unique characteristics, such as the effect of coherent teaching strategies (Perkins & Simmons, 1988).

An integrated curriculum's teaching needs to cover a broad knowledge according to its original design. However, most teachers possess only one or two academic backgrounds in four science subjects, making them uncomfortable or unable to teach integrated curricula (e.g. Harrell, 2010; Wang, 2018). Wan and Lee (2021) argued that the design of the textbook, curriculum, and syllabus should be consistent with the inner logic of that discipline. For example, as the only province in China that insists on implementing integrated science, 30 years of practice in the Zhejiang province have revealed a significant lesson: the deficiency of the coherence textbook that stems from the challenges in compiling a coherently integrated textbook (Wang, 2018).

Discrepancy between the effect of the two curricula

There is a considerable discrepancy in the research that has compared the effects of the two curricula (Venville et al., 2012). Some studies advocate for the integrated science curriculum due to its benefits to student science achievement, attitudes toward science, and scientific skills (such as Beichner et al., 1999; Czerniak & Johnson, 2015). Richardson and Showalter (1967) suggested that students who were delivered an integrated curriculum showed more interest in scientific problems and were more willing to apply their science knowledge in life by combining multiple subjects' knowledge facts than students in a separated science system. According to a meta-analysis conducted by Hurley (2001), an integrated curriculum has been shown to positively impact students in terms of discipline, attitudes, and engagement.

Conversely, several studies found that integrated science did not contribute to students' learning and attitudes, and integrated science teachers are often not qualified to teach such curricula (e.g. Harrell, 2010; Sun et al., 2014). Ross and Hogaboam-Gray (1998) found that students form a more solid knowledge structure in separated science education than students in integrated ones. A branch of researchers believes that separated disciplines are essential to teaching and studying because the segmented curriculum provides a sense of order that could help students understand themselves and their surroundings (Gardner, 2004; Schoenfeld, 2004). Young (2008) argued that segmented discipline is the price we have to pay to understand novel knowledge better.

Lee and Wan (2022) pointed out that, under ideal conditions, integrated science helps students to understand, master, and apply twenty-first-century competencies. However, we must not overlook the very prerequisite for this argument, that is, under ideal conditions. It is challenging to determine the reasons behind the effectiveness of the different curricula above. This challenge arises from the insufficient implementation details reported in existing studies. For example, Turpin and Cage (2004) conducted an experiment to compare the effects of integrated and traditional separated science

curricula on students' achievement, skills, and science learning attitudes and found that integrated science helps students more. These studies report limited information about their integrated science curriculum, such as the teacher's academic background, teaching strategies, and the textbook used. Also, studies that use interviews to investigate the effect of science curricula may carry subjectivity, and their results must be interpreted cautiously.

Commonly used methods in existing studies

The existing studies that investigated the impact of integrated science on student learning often employ a qualitative or mixed-research design (e.g. Sun et al., 2014; Wei, 2018). These studies typically use questionnaires to assess teachers' and students' reactions to integrated science, or they undertake experiments to evaluate the effect of integrated science. One of the disadvantages of these traditional methods is the small sample size, which hinders the generalizability of the results, especially when using convenient sampling (e.g. Zhang & He, 2012).

Recently, many studies analyzed this topic using the educational data mining paradigm. The decision tree is one of the most frequently adopted algorithms in educational data mining studies that are based on large-scale assessment data (Abu Saa et al., 2019; Gamazo & Martínez-Abad, 2020). Other commonly used methods include clustering, neural networks, logistic regression, support vector machines, and Naive Bayes. Thus, it is necessary to explore which algorithm performs the best among these widely used algorithms in the context of this study.

Theoretical framework

The present study used the socio-ecological theory as the foundational framework to guide the research design, which stresses social interaction in explaining learning and captures the complexity of factors that mould human development (Boyd & Bee, 2015).

Socio-ecological theory

The socio-ecological theory was first introduced as an ecological theory (Bronfenbrenner, 1979, 1986) that explains how factors and people in the environment impact an individual's development, emphasising the role of socialisation. Subsequently, Bronfenbrenner and Evans (2000) recognised the role of the individual in this theory and renamed it as the bioecological model, although some educational studies still refer to this theory as socio-ecological theory (e.g. Allen et al., 2018; Wang et al., 2023). Specifically, the socio-ecological theory is visualised as three concentric circles of systems or layers: microsystem, exosystem, and macrosystem.

The process-person-context-time (PPCT) framework

Bronfenbrenner and Morris (1998) introduced the Process-Person-Context-Time (PPCT) PPCT model based on the socio-ecological theory. The *process* dimension is similar to the original microsystem in the socio-ecological theory, including the

immediate interaction between the individual and the environment (Bronfenbrenner & Morris, 1998). The definition of the *person* dimension is also the same as the socio-ecological theory, indicating individual-related factors. Moreover, the *context* contains the higher-level environmental factors (Dai et al., 2023). Finally, the *time* dimension suggests when the interactions happen (Bronfenbrenner & Morris, 1998).

The socio-ecological theory is widely used as the theoretical framework in educational studies due to its capability to describe complex factors in the learning process. For example, Wang et al. (2023) also matched the socio-ecological model with variables from TIMSS 2019 and explored the high performances of East Asian students in mathematics based on machine learning algorithms. Dai et al. (2023) examined student reading achievement based on the PPCT model for the Programme for International Student Assessment (PISA) 2018 data.

Consistent with some examples illustrated above, this study also plans to analyze the large-scale assessment data. Considering the complexity of the variables in science learning, this study adopted the socio-ecological theory as the theoretical framework, particularly using the PPCT model to understand the relationships between science curriculum setting and student science learning experiences and outcomes.

Aligning the PPCT model to this study

This section aims to provide an alignment of the variables used in the present study to the PPCT model. However, the present study does not consider the *time* dimension of the PPCT model, given that the implementation of TIMSS 2019 does not contain several time periods.

First, the *person* dimension corresponds to the student-related variables. Dai et al. (2023) align the PPCT model with the PISA 2018 reading assessment and analytical framework, suggesting that the *person* dimension should contain student background and metacognitive variables, such as family background and attitudes toward learning. Given the similarities between PISA and TIMSS assessments, this study decided to follow the suggestions from Dai et al. (2023). Therefore, this study included four attitude variables (the first four variables at the bottom of Figure 1), two background variables (the fifth and sixth variables at the bottom of Figure 1), and the dependent variable of this study (student science achievement).

According to Bronfenbrenner and Morris (1998) and Dai et al. (2023), the *process* dimension ought to represent the interactions between individual and immediate surrounding environments, particularly in the context of education, encompassing variables related to teaching and learning. Because the science curriculum choice influences practical instruction, this study selected eight variables for this dimension, as shown on the right side of Figure 1: variables 4–8 were reported by teachers, whereas the remaining variables were reported by students.

The *context* dimension corresponds to the higher-level settings, such as variables related to school policy and governance (Dai et al., 2023). Although Dai et al. (2023) mentioned the curriculum in the *process* dimension, they also argued that the curriculum setting impacts student reading achievement in the *context* dimension. The decision between separated and integrated science curricula has several consequences, such as the recruitment of teachers, the choice of textbooks, and the design of educational

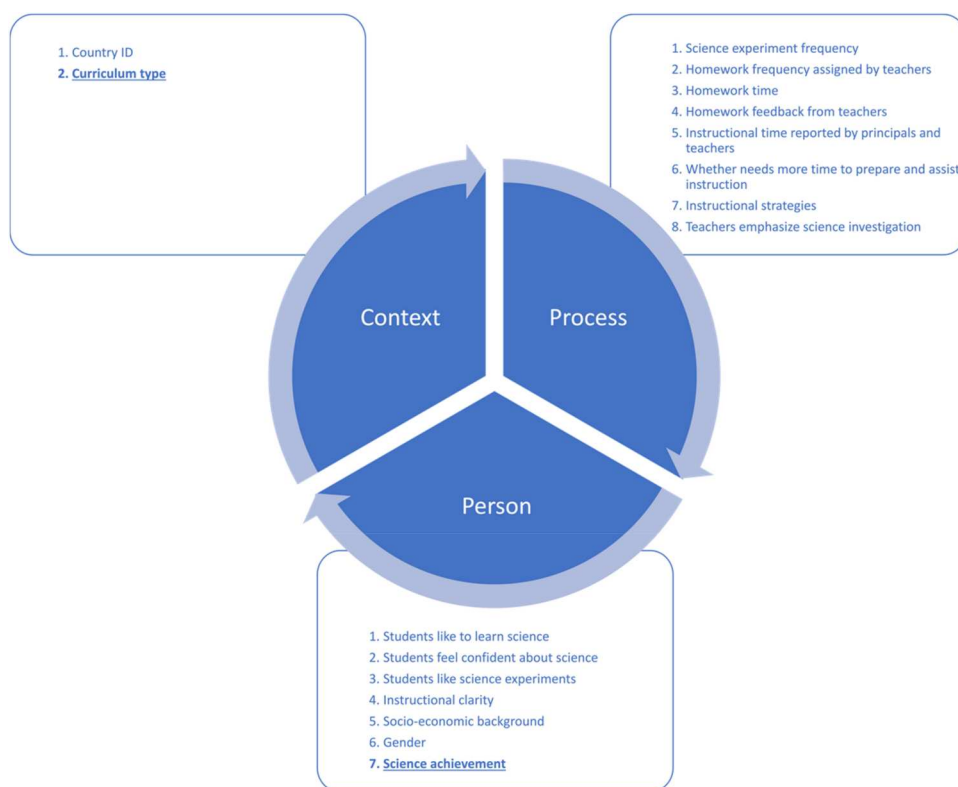


Figure 1. Theoretical model underlying the relationships of the independent and dependent variables with the covariates.

standards. Therefore, we grouped Country ID and Curriculum type into the *context* dimension for this study, as shown on the left side of Figure 1.

This study assumes that the setting of the science curriculum would directly influence student academic performance, which constitutes the main relationship this study aims to explore. These two variables of interest, science curriculum and student academic performance, are underlined in Figure 1.

Methodology

Data sources, participants, and procedures

This study employed the TIMSS 2019 Grade 8 Science data set (TIMSS Data, 2023). TIMSS measures the mathematics and science achievement of Grade 4 and 8 students from around 60 countries or regions worldwide. TIMSS collects curriculum information about each participant's country or region, which enables researchers to learn about curriculum instruction worldwide and monitor and compare the effect of educational policies (Wagemaker, 2002). In the TIMSS data set, the variable ITCOURSE denotes the 'Curriculum Type' (Table 1), and it provides information on the science curriculum (i.e. whether it is separated or integrated) taught in each participating country or region. The values of ITCOURSE include Physics = 2, Biology = 3, Chemistry = 4, Earth Science = 5, Integrated

Table 1. Independent variables used in the present study split by integrated and separated.

Variables	Integrated	Separated
Independent	Students like to learn science (BSBGSLI)	Students like to learn Biology/ Chemistry/ Physics/ Earth Science (BSBGSLB, BSBGSLC, BSBGSLP, BSBGSLE)
	Students feel confident about science (BSBGSCS)	Students feel confident about Biology/ Chemistry/ Physics/ Earth Science (BSBGSCB, BSBGSCC, BSBGSCP, BSBGSCE)
	Science experiment frequency (BSBS21)	Experiment frequency of Biology/ Chemistry/ Physics/ Earth science (BSBB22, BSBC32, BSBP37, BSBE27)
	Students like science experiments (BSBS22H)	Students like Biology/ Chemistry/ Physics/ Earth Science experiments (BSBB23H, BSBC33H, BSBP38H, BSBE28H)
	The Same Variables for Integrated and Separated Science Curriculum	
	Curriculum type (ITCOURSE)	
	Country ID (IDCNTRY)	
	Social-economic background (BSBGHER)	
	Homework frequency assigned by teachers (BTBS18A)	
	Homework time (BTBS18B)	
	Homework feedback from teachers (BTBS18CA, BTBS18CB, BTBS18CC, BTBS18CD, BTBS18CE)	
	Instructional clarity (BSBGICS)	
	Instructional time reported by principals and teachers (BTBS14, BCBG06B)	
	Whether needs more time to prepare and assist instruction (BTBG09D, BTBG09E)	
	Instructional strategies (BTBG12A, BTBG12B, BTBG12C, BTBG12D, BTBG12E, BTBG12F, BTBG12G)	
	Teachers emphasise science investigation (BTBSESI)	
	Gender (ITSEX)	
	Science teacher weight (SCIWGT)	

Note: The TIMSS code name for each variable is included in parentheses. The variable of Instructional strategies includes frequencies of linking instruction to daily lives, asking students to explain their answers, asking students to complete challenging exercises, encouraging classroom discussions, linking new content to prior knowledge, asking students to find their problem-solving procedures, and encouraging students to express their ideas. The variable of Homework feedback from teachers includes frequencies of correcting assignments, having students correct their homework, discussing the homework in class, monitoring whether the homework was completed, and using the homework to contribute to students' grades.

Science = 6, and other country-specific science courses = 10, 11, or 12. Thus, according to the classification of the science curriculum in the TIMSS 2019 results report (Mullis et al., 2020), the current study classified students whose ITCOURSE value was equal to six as belonging to the integrated science group, and the other students (i.e. ITCOURSE values of 2, 3, 4, or 5) as belonging to the separated science group. Therefore, **the large-scale data afforded by TIMSS about students, teachers, and curriculum-related data is ideally suited to the research questions posed in this study.**

Regarding data pre-processing, the IEA IDB Analyzer (IEA, 2022) and SPSS were used to extract the variables of interest and merge the raw data. After data cleaning, 309,511 entries from 44 countries remained. Appendix A details the missing data procedures. The code of this study can be found in a public online repository (Song, 2023).

Measures

The target (i.e. dependent) variable of this project was the student science achievement score. Following the procedures of previous studies, all plausible values (PVs) were used, and separate analyses were conducted for each PV.

The related literature suggests that the science curriculum type influences students' attitudes toward science. Therefore, **this study used two student attitude features as independent variables for students who were taught integrated curricula: liking to learn science and feeling confident about science.** The corresponding variables were selected for students taught using separate curricula, including liking and feeling confident

about Biology, Chemistry, Physics, and Earth Science. TIMSS standardised the original Likert-response items onto a common scale (Martin et al., 2020). Consequently, variables of separated curricula were averaged to enable a comparison between the two curriculum types. As an illustration, the liking variables for Biology, Chemistry, Physics, and Earth Science were averaged into a single variable. Other variables related to the instruction details include instructional clarity and time, experiment frequency, homework frequency, homework time, instruction preparation time needed by teachers, instructional strategies, emphasis on scientific investigation, and curriculum type.

TIMSS provides the sampling weights that allow an accurate estimate of the population characteristics by a sample of students (Martin et al., 2020). TIMSS provides three teacher weight variables to help analyze teacher context data at the student level, which are calculated by dividing the student weight by the number of that student's teachers (Fishbein et al., 2021). Moreover, it is also suggested to use subject-wise (e.g. science curriculum type) teacher weight when investigating teacher-level variables as student attributes (Liou & Hung, 2015). Specifically, some researchers believe that

if teachers of a particular subject (either math or science) in TIMSS are of analytic interest, it is then suitable to select either math teacher weight for analyses that include math teacher variables or science teacher weight for analyses that include science teacher variables. (Rutkowski et al., 2010, p. 144)

As variables four through eight in the Process dimension of Figure 1 were reported by science teachers, the science teacher's weight was used in the current study to decrease the sampling bias, given that the focus of this study was to understand the effect of the science curriculum type on student science achievement from an instructional perspective. In this respect, the science teacher's weight can provide a more accurate measure of instructional details than a student weight. Moreover, TIMSS data was released by country and region, and the IEA IDB Analyzer (IEA, 2022) was suggested as a tool to combine data across countries and regions (Fishbein et al., 2021). When using the IEA IDB Analyzer, if one teacher-level variable was selected, the teacher-level weight was automatically exported along with the selected variables. The independent variables employed in this study are shown in Table 1. The dependent variable is the plausible value of science achievement (BSSSCI01, BSSSCI02, BSSSCI03, BSSSCI04, BSSSCI05), the same for the separated and integrated groups.

How do the effects of the two curricula on students' achievement compare using the TIMSS 2019?

Randomly assigning students to one of two curriculum groups could guarantee a fair comparison between the curricula. When the random assignment is challenging to fulfill, propensity score matching is a promising alternative to balance the covariates' participant differences. The propensity score matching method could provide two matched groups on covariates, making it suitable to compare the effects of two curricula. Rosenbaum and Rubin (1983) proposed the propensity score method, which treats the confounding variables as covariates and estimates the conditional probability of each sample being classified into the treatment group on a group of covariates. The estimated probability is denoted as the propensity score, which could balance the covariate differences brought by selection bias.

Furthermore, Autenrieth et al. (2021) indicated that machine learning-based propensity score calculation methods could improve the estimation performance compared to commonly used logistic models. They proposed a new method based on stacked ensemble learning and achieved high accuracy. The stacked ensemble learning-based method demonstrated superior performance over logistic regression in three simulation studies, exhibiting less bias (average bias = 1.06%) in estimating treatment effects compared to logistic regression (average bias = 4.82%). Bias refers to the difference between the estimated and true treatment effects. Henceforth, the present study used stacked ensemble learning to calculate the propensity score.

Stacked ensemble learning employs a two-level analysis. First, it uses several base learners to predict the original data, and then it uses a meta-learner to predict a new data set comprising base learner outcomes (Autenrieth et al., 2021). This project selected KNN, random forest, and Naïve Bayes as stacking classifiers or base learners for stacked ensemble learning-based propensity score calculation and logistic regression as a meta classifier. After deleting replication data entries, imputing the missing data, dummy coding, and standardisation of the features, we split the whole data set into a training set (70% of the original data set) and a test set (30% of the original data set). The hyperparameter tuning procedure is detailed in Table A4.

The core idea behind the propensity score matching method is to match students who received an integrated science curriculum with similar students who received a separated science curriculum, based on the calculated propensity scores. First, stacked ensemble learning was used to estimate the propensity score for each student, considering all observed covariates. This study treated all independent variables, except for the curriculum type variable, as covariates. Second, students in the integrated curriculum group were matched with those in the separated curriculum group using the calculated propensity scores, employing the k -nearest neighbour (KNN) algorithm for this purpose. Third, the distributions of covariates for the two curriculum types were compared to ascertain if they were similar for both groups. Fourth, the effect of the science curriculum type was evaluated after obtaining balanced samples to understand the effect of curriculum type on each PV. The propensity score matching was also used to investigate the effect of inquiry teaching on student science achievement and attitudes based on the PSIA 2006 data, where five levels of inquiry teaching in estimating the propensity score were used (Jiang & McComas, 2015).

Does science curriculum type predict students' achievement in science?

This project employed linear regression, lasso regression, decision tree, random forest, and neural network to facilitate prediction interpretation and accuracy. The hyperparameter setting for each algorithm is similar to the above propensity score estimation. Table A5 shows the results of the hyperparameter setting and tuning for each PV. The obtained RMSE and R^2 for fitting with each PV were averaged to obtain the final performance of each algorithm.

Performance metrics

We used RMSE (Root Mean Square Error) and R^2 to evaluate the performance of the algorithms, where n denotes the sample size, y_i denotes the true dependent variable, $\hat{y}_i$ denotes

the estimated dependent variable, and $\bar{y}_i$ denotes the mean of the dependent variable.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y}_i)^2} \quad (2)$$

Results

Descriptive statistics

Table 2 shows the descriptive statistics of the cleaned data. The original questionnaire items are presented in Appendix A. Due to all *t*-test results violating the homogeneity of variance assumption (i.e. Levene's test results were all significant), this study used the non-parametric Mann–Whitney *U* test to compare the medians of variables between integrated and separated curricula. Hence, Table 2 summarises the medians for all the variables and means for the continuous variables. It also shows that the two curriculum types significantly differed on many *process* and *person* dimensional variables in the PPCT model (Figure 1), such as science experiment frequency and participant gender. Moreover, compared to students who received separated science curricula, those taking the integrated science curricula reported higher levels of instruction quality, greater confidence, and enjoyment in science learning. The instructor and principal average reported teaching time was higher in integrated than in separated curriculum implementations.

Table 3 shows the descriptive statistics for the categorical data, displaying response outcomes for each variable in descending order based on the percentage. Except for the homework frequency variable, for other frequency-related variables, the highest value (e.g. 3 or 4) corresponded to the lowest percentage, which stands for 'Never or rarely,' as indicated in Table 3.

Figure 2 shows the 30 most relevant and significant correlations between variables to guarantee readability, considering that the complete correlation matrix involving all variables is difficult to visualise. The abbreviations for each variable are shown in Table A1 in Appendix A. As the correlations of science achievement with the other variables were similar for each PV, Figure 2 only shows the correlations of the first PV with the other variables for simplicity. Figure A10 in Appendix A shows the correlations of each PV with the other variables grouped by two science curricula, respectively, consistent with Figure 2. The results show a medium correlation between variables related to instructional strategies. For example, the correlation coefficient between the 'Frequency of encouraging classroom discussions' and 'Frequency of asking students to complete challenging exercises' was 0.44, and the correlations of 'Frequency of linking instruction to daily lives' with 'Frequency of asking students to explain their answers' and 'Frequency of encouraging students to express their ideas' were both 0.30. The relationships between 'Teachers emphasize science investigation' and many instructional strategies variables were negative. For instance, the correlation between 'Teachers emphasize science investigation' and 'Frequency of asking students to complete challenging exercises' was -0.36 .

Table 2. The mann-whitney U test for all variables.

Variables	I_mean	S_mean	I_median	S_median	Mann-Whitney U	p-value	Effect Size
Frequency of asking students to complete challenging exercise	NA	NA	2	3	8.65e + 09	0.00	0.18
Instructional clarity	10.12	9.99	10.31	9.90	1.10e + 10	0.00	0.05
Frequency of encouraging classroom discussions	NA	NA	1	2	8.24e + 09	0.00	0.22
Students feel confident about science	10.14	10.00	9.76	9.74	1.06e + 10	0.00	0.01
Frequency of linking instruction to daily lives	NA	NA	1	1	1.13e + 10	0.00	0.07
Frequency of discussing homework in class	NA	NA	1	1	1.14e + 10	0.00	0.08
Teachers emphasise science investigation	10.19	9.21	10.24	8.82	1.34e + 10	0.00	0.28
Science experiments frequency	NA	NA	2	3	7.48e + 09	0.00	0.29
Students like science experiments	NA	NA	1	2	6.94e + 09	0.00	0.34
Frequency of correcting assignments	NA	NA	1	1	1.16e + 10	0.00	0.10
Frequency of encouraging students to express their ideas	NA	NA	1	1	1.07e + 10	0.00	0.02
Gender	NA	NA	1	2	1.04e + 10	0.00	0.01
Homework frequency assigned by teachers	NA	NA	3	3	1.01e + 10	0.00	0.04
Homework time	NA	NA	2	2	1.11e + 10	0.00	0.05
Instruction time reported by principals	339.79	294.21	336	280	1.69e + 10	0.00	0.60
Instructional time reported by teachers	188.93	90.59	180	90	1.91e + 10	0.00	0.82
Students like to learn science	10.19	9.90	9.82	9.76	1.11e + 10	0.00	0.06
Frequency of monitoring whether the homework was completed	NA	NA	1	1	1.06e + 10	0.00	0.01
Whether needs more time to assist with instruction	NA	NA	1	2	8.63e + 09	0.00	0.18
Whether needs more time to prepare instruction	NA	NA	2	2	9.81e + 09	0.00	0.07
Frequency of linking new content to prior knowledge	NA	NA	1	1	1.24e + 10	0.00	0.18
Frequency of asking students to find their problem-solving procedures	NA	NA	2	2	1.12e + 10	0.00	0.06
The first plausible values of science achievement	486.16	515.31	494.73	519.78	9.01e + 09	0.00	0.14
The second plausible values of science achievement	486.35	515.34	494.88	519.78	9.01e + 09	0.00	0.14
The third plausible values of science achievement	487.59	516.31	495.58	520.69	9.02e + 09	0.00	0.14
The fourth plausible values of science achievement	485.91	515.22	494.25	519.12	9.00e + 09	0.00	0.14
The fifth plausible values of science achievement	486.69	515.68	495.31	519.71	9.02e + 09	0.00	0.14
SES (socio-economic background)	10.24	10.81	10.24	10.84	8.37e + 09	0.00	0.21
Frequency of having students correct their homework	NA	NA	2	2	1.10e + 10	0.00	0.05
Frequency of correcting assignments	NA	NA	1	1	9.35e + 09	0.00	0.11
Frequency of using homework to contribute to students' grades	NA	NA	2	2	1.20e + 10	0.00	0.15

Note: I_mean and I_median denote the mean and median values, respectively, for the integrated curriculum variables; S_mean and S_median denote the mean and median values, respectively, for the separated curriculum variables.

The strongest association in Figure 2 was between 'students like to learn science' and 'students feel confident about science' ($r = 0.69$). Student gender was not associated with any other variable. Meanwhile, a strong positive relationship was found between student science achievement and socio-economic background ($r = 0.44$).

Table 3. Descriptive statistics for categorical variables.

Variable	Outcome	Percent	Variable	Outcome	Percent
Frequency of discussing	1	57.04	Frequency of using homework to contribute to students' grades	2	42.05
	2	25.41		1	40.26
	3	17.54		3	17.69
Frequency of having students correct their homework	2	57.04	Frequency of monitoring whether the homework was completed	1	80.87
	1	25.41		2	18.27
	3	17.54		3	0.86
Frequency of correcting assignments	1	65.87	Science experiment frequency	2	36.95
	2	30.88		3	29.72
	3	3.25		1	21.38
Frequency of encouraging students to express their ideas				4	11.95
	1	64.98	Frequency of asking students to explain their answers	1	62.83
	2	23.93		2	26.29
	3	10.85		3	10.69
Students like science experiments	4	0.24		4	0.19
	1	46.22	Whether needs more time to prepare instruction	2	44.53
	2	38.36		1	24.16
	3	10.74		3	23.68
Whether needs more time to assist with instruction	4	4.68		4	7.63
	1	45.65	Frequency of asking students to find their problem-solving procedures	2	37.33
	2	42.25		1	31.21
	3	9.81		3	29.3
Frequency of asking students to complete challenging exercises	4	2.29		4	2.16
	3	38.79	Frequency of encouraging classroom discussions	1	46.1
	2	34.8		2	29.05
	1	22.97		3	23.5
Frequency of linking instruction to daily lives	4	3.44		4	1.36
	1	62.57	Frequency of linking new content to prior knowledge	1	73.55
	2	26.16		2	20.63
	3	11.14		3	5.73
Homework frequency	4	0.12		4	0.09
	3	51.7	Homework time	2	52.26
	2	20.56		1	32.06
	4	10.99		3	14.05
Gender	1	8.95		4	1.05
	5	7.8	Curriculum type	5	0.57
	1	50.56		0	67.43
	2	49.46		1	32.57

Note: The interpretation of each response outcome is provided in the questionnaire in Figures A3–A9 in [Appendix A](#). The response options for each variable are coded from left to right in ascending numerical order. In terms of the 'Gender' variable, '1' corresponds to male, and '2' corresponds to female. In terms of the 'Curriculum type' variable, '0' represents an integrated curriculum, and '1' represents a separated curriculum.

How do the effects of the two curricula on student achievement compare in TIMSS 2019?

The final propensity score prediction accuracy on the test set was 0.9998. The left panel of [Figure 3](#) shows the density of propensity score logit before matching, in which the two curricula have very different density distributions, so the comparison of the two programmes based on this will be unfair. Therefore, we applied propensity score matching with the first PV as the outcome. The right panel of [Figure 3](#) shows that, now, the integrated and separated groups have nearly the same density distribution, enabling a fair comparison between the two curricula. The pattern in [Figure 3](#) persisted in the matching with all PVs. Only the results for the first PV were presented in this section, and the results of other PVs were presented in Figures A11–A14 in [Appendix A](#).

[Figure 4](#) shows the overview of 'students like to learn science,' 'students feel confident about science,' and 'instructional clarity' before and after matching with the first PV,

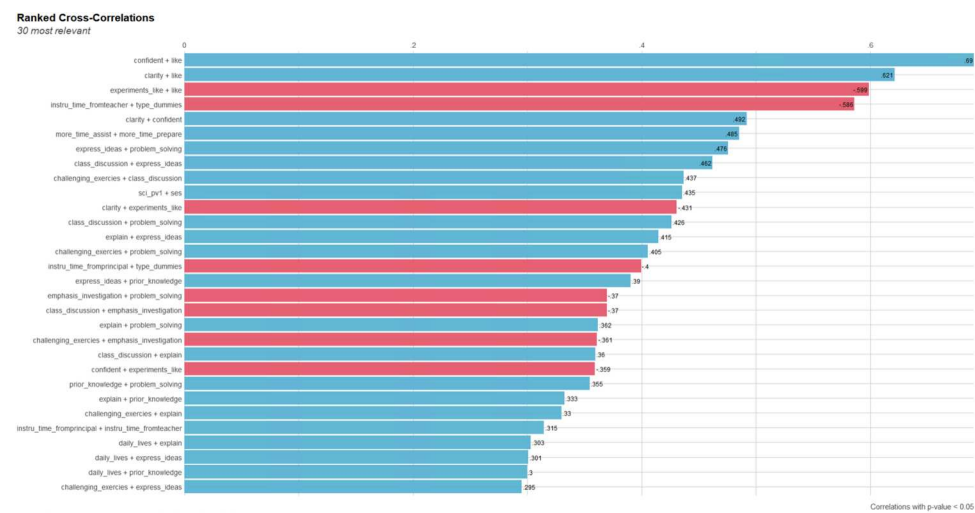


Figure 2. The thirty most relevant correlations between the independent and dependent variables.

respectively. Other PV results were presented in Figures A15–A18 in Appendix A. After matching, these three variable distributions mostly overlap between integrated and separated groups.

The average treatment effect is shown in Table 4. As all the data did not satisfy the normality assumption, the non-parametric Mann–Whitney U test was conducted to compare the two types of science curricula for each PV. All the Mann–Whitney U test results with Bonferroni correction indicate that the integrated group’s scientific achievement was significantly higher than that of the separated group after controlling for all

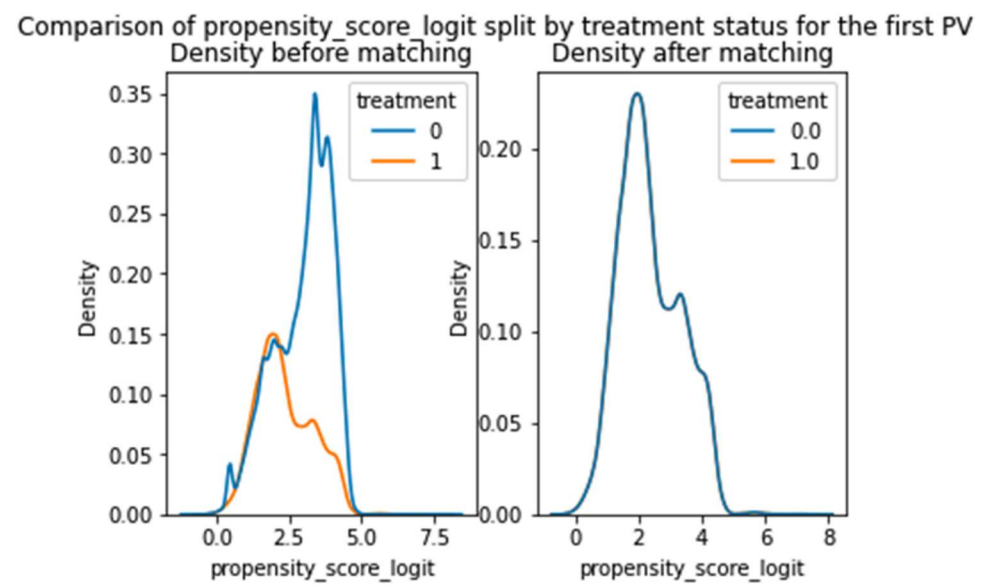


Figure 3. Propensity score logit before and after matching with the first PV.

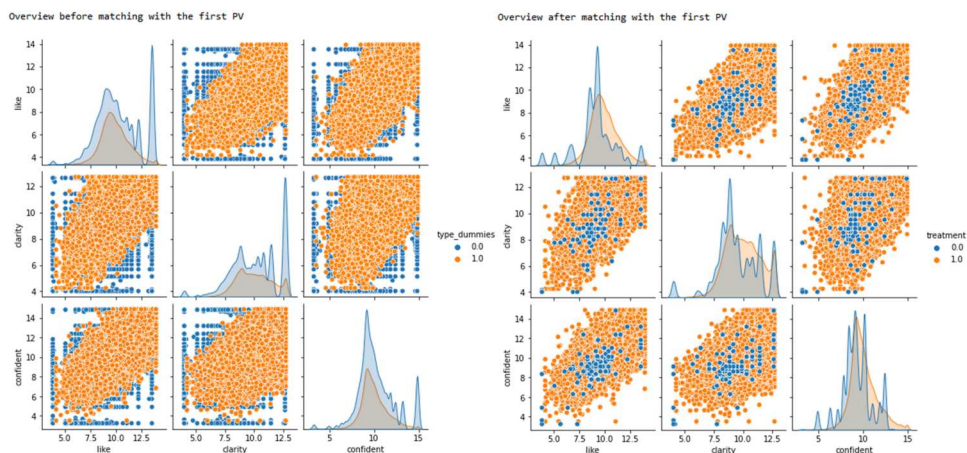


Figure 4. Overview of three covariates before and after matching with the first PV.

other covariates, as shown in Table 4. All Cohen’s *d* values were around 0.2, suggesting a small effect of the difference between the two groups. Thus, although the integrated curricula showed a slight advantage over the separated curricula, the effect was small.

Does science curriculum type predict students’ achievement in science?

Prediction results

Random forest outperformed the rest of the models employed in this study, as shown in Table 5. Linear and lasso regression performed the worst among the five algorithms, suggesting that the relationship between features and target was not linear in the context of this study.

Feature importance

Figure 5 shows the average feature importance coefficients obtained from the random forest with each PV, which all identified student socio-economic status as the most important predictor of achievement. Other essential features include ‘students feel confident about science,’ ‘instructional time,’ and ‘science experiment frequency’. The last two variables relate to instructional details, which provide the basis for future

Table 4. Students’ science achievement after propensity score matching.

Science plausible value	Treatment	Mean	Variance	Standard deviation	Mann-whitney <i>U</i>	<i>p</i> -value	Cohen’s <i>d</i>
PV1	Integrated	546.62	6133.17	78.31	3.97e + 09	0.00	0.22
	Separated	515.68	7855.70	88.63			
PV2	Integrated	544.61	4854.52	69.67	4.05e + 09	0.00	0.20
	Separated	515.34	7795.92	88.29			
PV3	Integrated	552.33	6774.32	82.31	3.86e + 09	0.00	0.24
	Separated	516.31	7749.86	88.03			
PV4	Integrated	546.29	5703.52	75.52	3.93e + 09	0.00	0.23
	Separated	515.21	7872.63	88.73			
PV5	Integrated	546.62	6133.17	78.31	4.09e + 09	0.00	0.20
	Separated	515.68	7855.70	88.63			

Table 5. Comparison of the prediction results.

Algorithms	R^2			RMSE		
	Training	Evaluation	Testing	Training	Evaluation	Testing
Linear regression	0.21	N/A	0.21	94.42	N/A	94.43
Lasso regression	0.22	0.22	0.22	94.24	94.26	94.24
Decision tree	0.31	0.29	0.29	88.65	89.59	89.88
Random forest	0.73	0.50	0.51	55.01	75.48	74.92
Neural networks	0.22	0.22	0.22	94.33	94.01	94.34

Note: N/A = not applicable, as there are no hyperparameters to tune for linear regression.

curriculum design and revision. Although the curriculum type coefficient was non-zero, curriculum type was also not the most critical feature according to these findings, suggesting that the contribution of curriculum type in directly predicting students' achievement was minimal. Overall, the prediction results showed that the curriculum type was a weak predictor of student science performance, which is supported by the average coefficients of each predictor in Table 6.

Random forest explained the most variance of student science achievement among all algorithms used in this study, warranting a deeper investigation of this algorithm. We selected two random students to explain the decision process of the random forest algorithm fitting with the first PV and investigated the most critical features in this decision process, as shown in Tables A2 and A3 in Appendix A, respectively.

For example, the decision process for Student 31 is Prediction: 530.26 = 496.86 (train- ing set mean) – 19.01 (loss from socio-economic background) + 14.34 (gain from

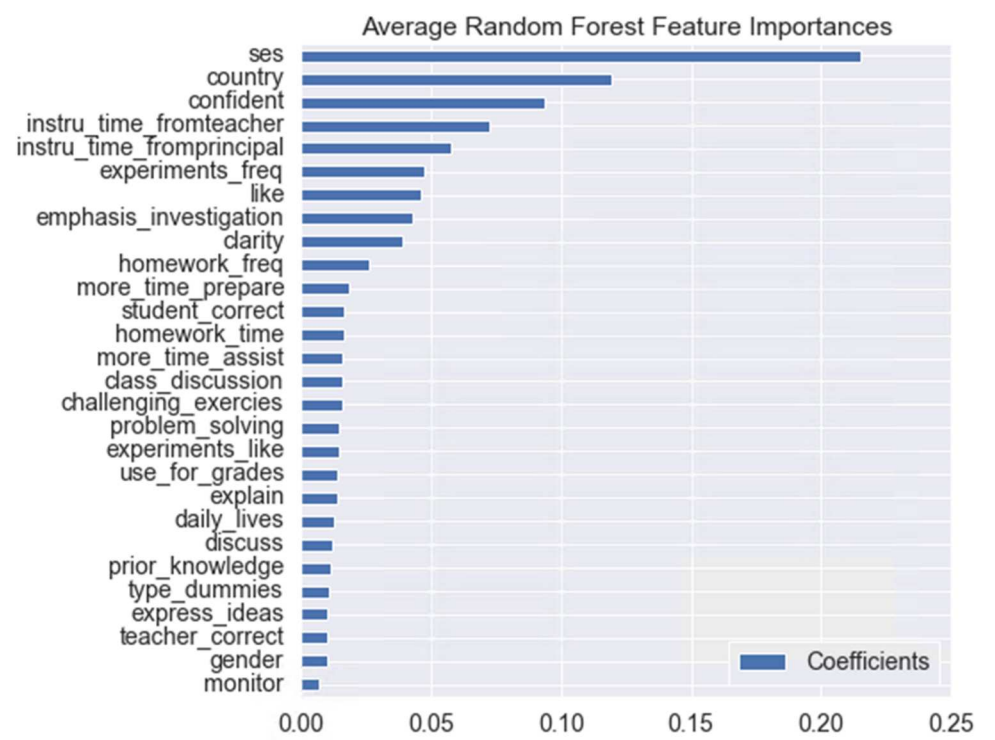


Figure 5. Feature importance.

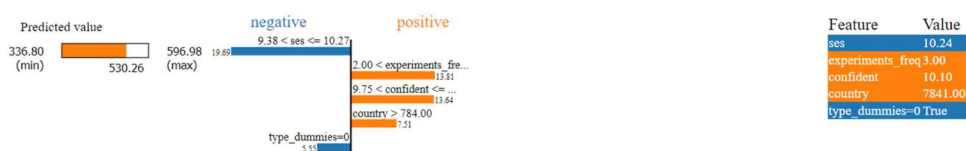
Table 6. Average variable coefficients in the random forest.

Variable	Coefficients
SES (socio-economic background)	0.215
Country	0.119
Students feel confident about science	0.094
Instructional time reported by teachers	0.072
Instruction time reported by principals	0.058
Science experiments frequency	0.047
Students like to learn science	0.046
Teachers emphasise science investigation	0.043
Instructional clarity	0.039
Homework frequency assigned by teachers	0.026
Whether needs more time to prepare instruction	0.018
Frequency of having students correct their own homework	0.017
Homework time	0.016
Whether needs more time to assist with instruction	0.016
Frequency of encouraging classroom discussions	0.016
Frequency of asking students to complete challenging exercises	0.016
Frequency of asking students to find their problem-solving procedures	0.015
Students like science experiments	0.015
Frequency of using homework to contribute to students' grades	0.014
Frequency of asking students to explain their answers	0.014
Frequency of linking instruction to daily lives	0.012
Frequency of discussing the homework in class	0.012
Frequency of linking new content to prior knowledge	0.011
Curriculum type	0.010
Frequency of encouraging students to express their ideas	0.010
Frequency of correcting assignments	0.010
Gender	0.010
Frequency of monitoring whether the homework was completed	0.007

teachers' reported instructional time) + 11.85 (gain from country identity) + 7.75 (gain from science experiment frequency) – 6.10 (loss from principals' reported instructional time) + 5.98 (gain from instructional clarity) + ... + $\beta_p X_p$.

Curriculum type played a minor role for students 31 and 3335, decreasing science prediction by 1.45 and 3.23, respectively, as indicated in Table A3.

Figures 6 and 7 summarise the five most essential features in the random forest's decision process for students 31 and 3335, respectively. Figure 6 suggests that student 31 performed well on the science assessment because three significant predictors increased this student's performance prediction. Combined with the feature importance yielded by the random forest shown in Figure 5, the five most important features in predicting student science achievement were 'students feel confident about science,' socio-economic status, country identities, and reported instructional time by teachers and principals. Among these five significant predictors, stakeholders can work on improving or reforming students' confidence in science and receiving instructional time.

**Figure 6.** Five most important nodes in the random forest decision process for student 31.

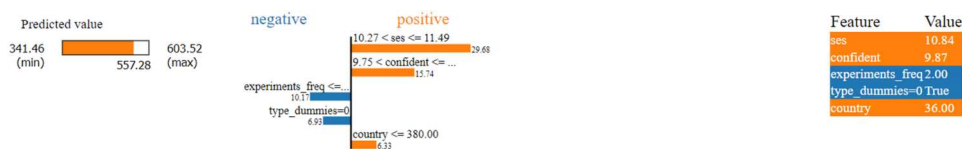


Figure 7. Five most important nodes in the random forest decision process for student 3335.

Interaction analysis

This section investigates whether the presence of the curriculum type influences the relationship between independent variables and the dependent variable. Figure 8 draws the relationships between curriculum type and student attitude variables, and Figure 9 depicts the relationship between curriculum type and instructional clarity. Trends in Figures 8 and 9 were observed in fitting models for each PV. Only the results from the first PV were presented in the current section, and the results for other PVs were presented in Figures A19–A26 in Appendix A. The overall trends of the variables ‘students feel confident about science’ and ‘instructional clarity’ were similar for the two curriculum types.

However, according to the right panel of Figure 8, at a horizontal axis value of 8, the estimated SHAP (Shapley Additive exPlanations) value for ‘Students like to learn science’ was larger than 0 for the separated group (red), indicating that students’ achievement increased the more they liked science when students displayed low or medium enjoyment in learning science. Conversely, for the integrated group (blue), the SHAP value was smaller than 0 at this point (X-axis = 8), suggesting that an increase in liking science was associated with a decrease in science performance for these students. Moreover, the relationship was reversed when the students highly enjoyed science learning (e.g. X-axis = 14), suggesting that the curriculum type influenced the relationship between liking science and student achievement.

Discussion

The curriculum setting is critical in cultivating student science literacy by influencing the formation of student thinking habits, problem-solving strategies, knowledge concept structure, and studying attitudes (Beichner et al., 1999; Czerniak & Johnson, 2015; Lee

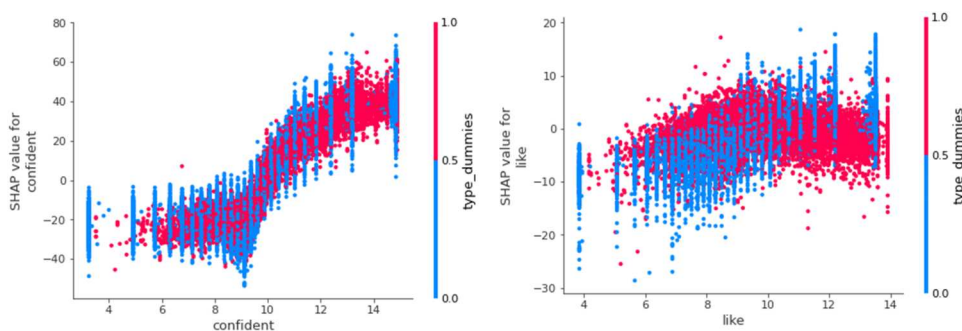


Figure 8. Interactions between student attitudes and curriculum type for the first PV.

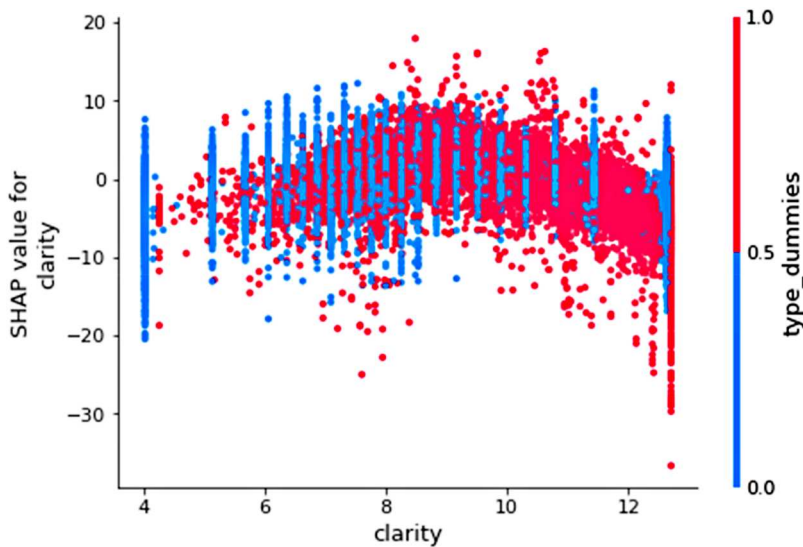


Figure 9. Interaction between instructional clarity and curriculum type for the first PV.

& Wan, 2022). Also, curriculum relates to science teacher demands, curriculum standards, educational measurements, and many other critical educational aspects. **Primary and secondary education constitute the foundation for the pursuit of higher education.** Thus, the decision of which curriculum type to use in a particular country or region is significant, as it can profoundly impact students' future educational trajectories.

Despite the importance of the curriculum setting, previous research results did not reach an agreement on which curriculum aids students more. One reason is that these studies often collect data from nearby schools, thus their results may only suit the selected school units. This questions the generalizability of these conclusions to nearby cities and, especially, to whole countries or regions. The integrated science curriculum is relatively new compared to the separated one. Therefore, its relatively short developmental and implemental history indicates that several fundamental challenges have yet to be solved. Several aspects must be considered, including a universal curriculum standard, a coherent textbook, a competent teacher, and a valid and reliable measurement instrument. When the implementation of the integrated curriculum does not match the design, the integrated curriculum is sometimes less helpful than the separated curriculum in some student attitudes and instructional details. However, insufficient information regarding curriculum implementation in existing studies hinders the determination of the exact reason for the poor effects of an integrated curriculum.

Studies examining one region's curriculum effect are likely to influence educational change in that region. However, the outcome is affected if these studies yield potentially biased results. Therefore, this study analyzed large-scale assessment data to avoid the drawbacks of the existing studies. The curriculum implementation details from almost all participants in TIMSS 2019 help to elucidate this issue. However, one premise of the analysis is that the TIMSS 2019 science assessment was not designed to favour either type of science curriculum. **Although the background questionnaires provide a**

different version for separated and integrated science curricula, the achievement items were designed for different science knowledge concepts based on the released item information from TIMSS 2019 (Fishbein et al., 2021). For example, the SP52021 item measures ecosystems in the biology content area in TIMSS 2019 (Fishbein et al., 2021).

This study calculated propensity scores based on stacked ensemble learning to compare the impact of two curricula on student science achievement. First, we used KNN to generate the matched integrated and separated groups. The Mann–Whitney *U* test results suggest that the integrated group slightly outperformed the separated one. However, the effect size was small, suggesting that the learning difference between the two curricula was not large among TIMSS participants.

Furthermore, this study investigated whether the science curriculum type is critical in predicting student academic performance by comparing multiple algorithms. Random forest achieved the best prediction accuracy among the five algorithms, with the possibility of overfitting. The poor performance of linear and lasso regression suggested that the relationship between independent and dependent variables was likely non-linear. Therefore, results suggested that the random forest algorithm could be suitable for similar tasks where the relationship between a large set of independent and dependent variables is likely to be non-linear.

Moreover, the first research question found that the effect of integrated curricula on students' achievement was better than that of separated ones. The machine learning results found that curriculum setting was a weak predictor in directly predicting students' achievement. Combining two results leads to the conclusion that curriculum setting may indirectly influence students' science literacy by influencing the covariates included in this study. The interaction analysis revealed that the science curriculum influenced the relationship between students' enjoyment of science learning and their achievement. Among the students in the separated group with low or medium enjoyment in learning science, a positive association existed between liking science and achievement. In contrast, within the integrated group, an increase in liking science corresponded to decreased science performance; furthermore, this relationship was reversed for students with high enjoyment. This interaction relationship indicated that the marginal advantage of an integrated science curriculum in enhancing student achievement was indirectly realised.

Moreover, the average value for most of the independent variables in Table 2 was higher for the integrated group (e.g. 'instructional clarity,' 'students like to learn science,' 'students feel confident about science,' and 'teachers emphasize science investigation'). Notably, the average instructional time reported by teachers who teach integrated science was almost twice that of teachers who teach separated science. These descriptive statistics constitute another piece of evidence that supports the indirect marginal advantage of integrated science over separated science curricula. Therefore, a follow-up analysis (e.g. structural equation modelling) is needed to examine the indirect influence mechanism between curriculum type, other variables in the PPCT model, and student achievement.

Educational implications

Many students who received integrated science curricula reported unclear instruction, which echoes the previous results that the implementation effect of the integrated

science curriculum requires several prerequisites. For example, a coherent textbook, a competent instructor, and an appropriate measurement instrument indicate that educational practitioners and researchers could devote themselves to achieving the above-mentioned prerequisites. Moreover, some instruction or curriculum-related background variables in the *process* dimension of the PPCT model (e.g. science experiment frequency) were critical in predicting student achievement. Concomitantly, many students disliked science experiments, although they may like science in general. Indeed, **the correlations between ‘students like science experiments’ and ‘students like to learn science,’ ‘instructional clarity,’ and ‘students feel confident about science’ were all significantly negative: -0.60 , -0.43 , and -0.36 , respectively.** This attitude is important in the *person* dimension of the PPCT model. Henceforth, improving the curriculum of science experiments instruction is also necessary.

Although the answers to the two research questions appear to be contradictory, propensity score matching revealed that integrated science benefited students more than separated curricula. However, **machine learning algorithms indicated that the science curricula did not predict student achievement directly.** It is worth noting that the findings are not in opposition to the discussion that follows. Machine learning algorithms revealed feature interaction relationships, combining the descriptive statistics, enabling the inference that the instructional details of integrated science curricula exhibited greater variance among all participating countries or regions than separated curricula. This disparity arises from the difficulties in satisfying the essential prerequisites of an integrated curriculum, as discussed previously.

It is possible that some implementations satisfy the prerequisites of an integrated science curriculum while others do not. This discrepancy leads to large variability in instructional detail variables. However, when only considering central tendency statistics, such as the mean of these instructional details, those differences were mitigated by the average. On average, students who received integrated science instruction showed more positive attitudes toward science learning enjoyment and confidence than those who received separated science instruction, as shown in Table 2. Therefore, it is critical for some educational systems to elucidate the reasons behind the poor direct effect of science curricula on student achievement.

Theoretical implications

Regarding the long-standing debate between separated and integrated science curricula, this study first suggested that an integrated curriculum benefited students marginally more than a separated one based on a large-scale data set. Additionally, several features predicting students’ performance were identified, including ‘students feel confident about science,’ ‘instructional time reported by teachers,’ and ‘science experiment frequency.’ The relationships between curriculum background features that predict student performance verified the several prerequisites for effective curriculum implementation. Specifically, when students reported explicit instruction, they often liked learning science and felt confident about science. However, when students performed positively on these three variables, they tended to dislike science experiments, indicating that the overall science experiments were not satisfactory for students, at least among TIMSS 2019 participants. Moreover, this study found that the relationship

between the learning process-related variables and student science performance was not linear. Future research is needed to verify whether curriculum type indirectly influences students' achievement through curriculum aspects, such as instructional time.

Methodological implications

Previous studies on this issue have primarily been empirical with small sample sizes. If the sample cannot represent the population or if the implementation does not meet the intended design, then the student benefits from the curriculum will be affected. To overcome the limitations of existing studies, this study employed a large-scale assessment data set and machine learning approaches to compare the effects of the curricula on student learning gains. The significant comparison results and high prediction suggested that machine learning methods are well suited to this research problem. Big educational data could answer questions that traditional methods could not quickly solve. Especially when the problem relates to educational policy or is of immediate worldwide interest, educational data mining techniques can be used to extract implicit relationships and meaning from big educational data. The current study employed the student science achievement as the dependent variable and all variables in the *Process* dimension of the PPCT model as the independent variables. However, using some variables in the *Process* dimension as the dependent variable would be worthwhile to explore in the future to better understand whether the curriculum type directly predicts such attitude variables (e.g. 'students feel confident about science'). Moreover, some variables in the *Process* dimension of the PPCT model were reported by the teacher. Therefore, in future analysis, applying the hierarchical model for teacher- and student-level variables is necessary to investigate whether multilevel relationships exist among variables.

Practical implications

The most salient predictive features include student socio-economic status, attitudes toward science, instructional time, and experiment conducting frequency. Although it is challenging to change some of these student features (e.g. socio-economic status), others have the potential to be impacted by practitioner-led interventions, situated in the *process* dimension of the PPCT model adapted for this study (e.g. changing the instructional time and experiment frequency based on the student's needs and capabilities).

The high positive correlation between 'students like to learn science,' 'students feel confident about science,' and 'instructional clarity' was in line with the previous conclusions. However, despite these positive associations, this study also found that students often dislike scientific experiments. Such a finding warrants further investigation because conducting experiments may teach students several knowledge and skills. Moreover, it is important to pay attention to student attitudes, as they may influence their learning experiences and outcomes.

Limitations

This study found that integrated science was marginally more beneficial to students than a separated curriculum. However, the present study did not explain why integrated

science benefits students more. To answer this question, exploring the internal influence mechanisms between curriculum type and student achievement is necessary, which is crucial for understanding and enhancing science curriculum implementations.

Moreover, this study did not analyze the specific participant country because the primary purpose was to determine which curriculum benefitted students more. However, it is worthwhile to deeply analyze participant countries and their curriculum-related policies, design, practice, and feedback from teachers and students to understand why integrated science benefits students more in one country than in another.

Conclusions

This study applied several popular machine learning algorithms to compare the effects of two curricula on student science learning. Two comparable groups were obtained using propensity score matching, and the integrated group showed slightly better science achievement than the separated group among TIMSS 2019 participants. Findings revealed that random forests achieved the best prediction accuracy. In addition to student socio-economic backgrounds, other essential features could be considered in future research, including student attitudes toward science and instruction-related background details. These relationships need to be examined to further understand the influence mechanism of curriculum setting on student science literacy.

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Ethics and data statement

This study uses the TIMSS publicly available data set available at <https://timss2019.org/international-database>. Therefore, no ethics requirements are needed.

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Appendix A

Data pre-processing

The raw data contains duplicates of data for one student in different classes, which means the same student taught by the same teacher but in different classes has two entries in the data set. Therefore, unique data records were kept for each student.

The missing data reports for integrated and separated curricula are shown in Figures A1 and A2, respectively. After checking the region sources of the five parallel blanks in Figure A2, we found that the data on variables linked to Earth Science for two countries, Lebanon and Sweden, in the separated curriculum was missing systematically. The *Lebanon* and *Sweden* chapters in the *TIMSS 2019 Encyclopedia* (Kelly et al., 2020) suggested that Earth Science and Biology are integrated in Lebanon (Center for Educational Research and Development, 2020, p. 6), whereas Earth Science and Physics are integrated in Sweden (Axelsson, 2020, p. 12). Hence, as it is not easy to determine the curriculum type partition for these two nations, the data from these two countries were excluded from this analysis.

Entries for variables that have less than 5% missing records were removed. Deleting data for variables with systematically missing patterns may lead to bias. In this case, the preferable option is to impute the missing data. Therefore, the present study employed the KNN imputation technique. As part of the pre-processing step, dummy coding was adopted to represent categorical variables, and the features were standardised for their use in similarity matching.

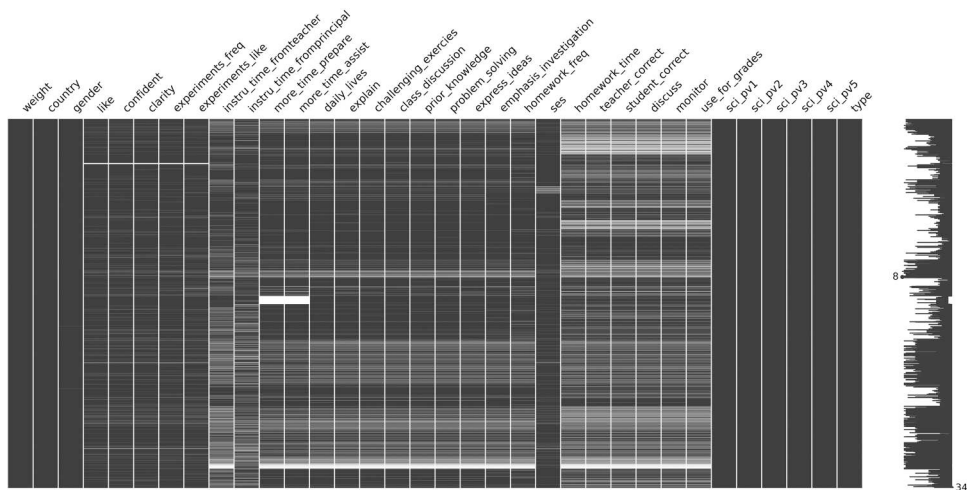


Figure A1. Missing data report for the integrated curriculum.
Note: Due to space restrictions, abbreviations are used for variable names. The specific variable for each abbreviation is shown in [Table A1](#) in Appendix A.

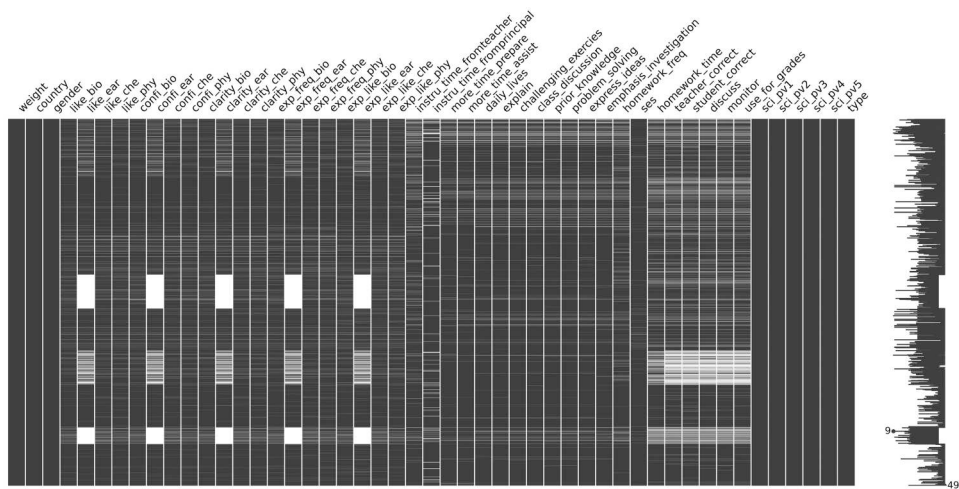


Figure A2. Missing data report for the separated curriculum.

How much do you agree with these statements about learning science?

Fill **one** circle for each line.

	Agree a lot	Agree a little	Disagree a little	Disagree a lot
a) I enjoy learning science	☐	☐	☐	☐
b) I wish I did not have to study science	☐	☐	☐	☐
c) Science is boring	☐	☐	☐	☐
d) I learn many interesting things in science	☐	☐	☐	☐
e) I like science	☐	☐	☐	☐
f) I look forward to learning science in school	☐	☐	☐	☐
g) Science teaches me how things in the world work	☐	☐	☐	☐
h) I like to conduct science experiments	☐	☐	☐	☐
i) Science is one of my favorite subjects	☐	☐	☐	☐

Figure A3. Students like learning science scale (IEA, 2018a).

How much do you agree with these statements about science?

*Fill **one** circle for each line.*

	Agree a lot	Agree a little	Disagree a little	Disagree a lot
a) I usually do well in science -----	☐	☐	☐	☐
b) Science is more difficult for me than for many of my classmates ----	☐	☐	☐	☐
c) Science is not one of my strengths -----	☐	☐	☐	☐
d) I learn things quickly in science -----	☐	☐	☐	☐
e) I am good at working out difficult science problems -----	☐	☐	☐	☐
f) My teacher tells me I am good at science -----	☐	☐	☐	☐
g) Science is harder for me than any other subject -----	☐	☐	☐	☐
h) Science makes me confused -----	☐	☐	☐	☐

Figure A4. Students feel confident about science scale (IEA, 2018a).

How much do you agree with these statements about science?

*Fill **one** circle for each line.*

	Agree a lot	Agree a little	Disagree a little	Disagree a lot
a) I think learning science will help me in my daily life	☐	☐	☐	☐
b) I need science to learn other school subjects	☐	☐	☐	☐
c) I need to do well in science to get into the <university> of my choice -	☐	☐	☐	☐
d) I need to do well in science to get the job I want	☐	☐	☐	☐
e) I would like a job that involves using science	☐	☐	☐	☐
f) It is important to learn about science to get ahead in the world	☐	☐	☐	☐
g) Learning science will give me more job opportunities when I am an adult	☐	☐	☐	☐
h) My parents think that it is important that I do well in science	☐	☐	☐	☐
i) It is important to do well in science	☐	☐	☐	☐

Figure A5. Students value science scale (IEA, 2018a).

How often do you do the following in teaching this class?

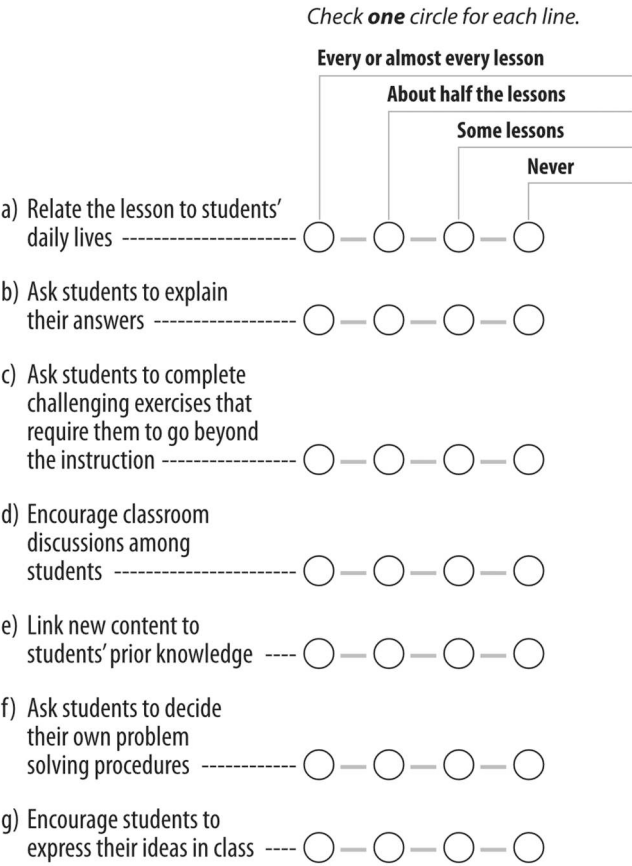


Figure A6. Science instructional practice scale (IEA, 2018b).

B. When your teacher gives you homework in the following subjects, about how many minutes do you usually spend on your homework?

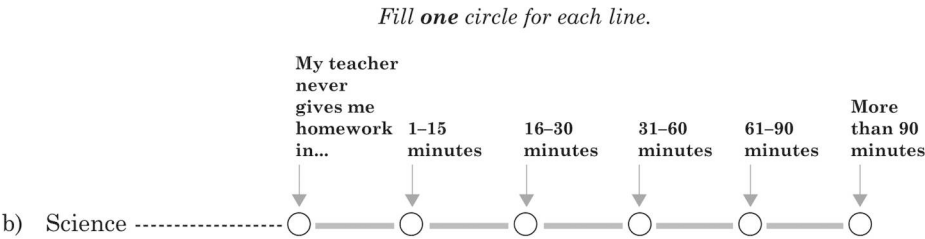


Figure A7. Time spent on science homework (IEA, 2018a).

C. How often do you do the following with the science homework assignments for this class?

Check **one** circle for each line.

	Always or almost always	Sometimes	Never or almost never
a) Correct assignments and give feedback to students -----	☐	☐	☐
b) Have students correct their own homework -----	☐	☐	☐
c) Discuss the homework in class -----	☐	☐	☐
d) Monitor whether or not the homework was completed ----	☐	☐	☐
e) Use the homework to contribute towards students' grades or marks -----	☐	☐	☐

Figure A8. Homework scale (IEA, 2018b).

Indicate the extent to which you agree or disagree with each of the following statements.

Check **one** circle for each line.

	Agree a lot	Agree a little	Disagree a little	Disagree a lot
d) I need more time to prepare for class -----	☐	☐	☐	☐
e) I need more time to assist individual students -----	☐	☐	☐	☐

Figure A9. Whether needs more time (IEA, 2018b).

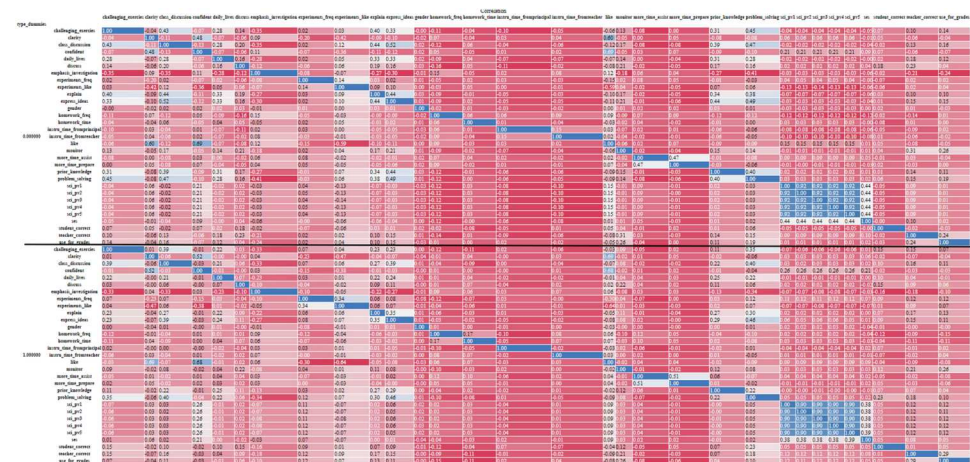


Figure A10. The correlations within the integrated and separated curricula, respectively.

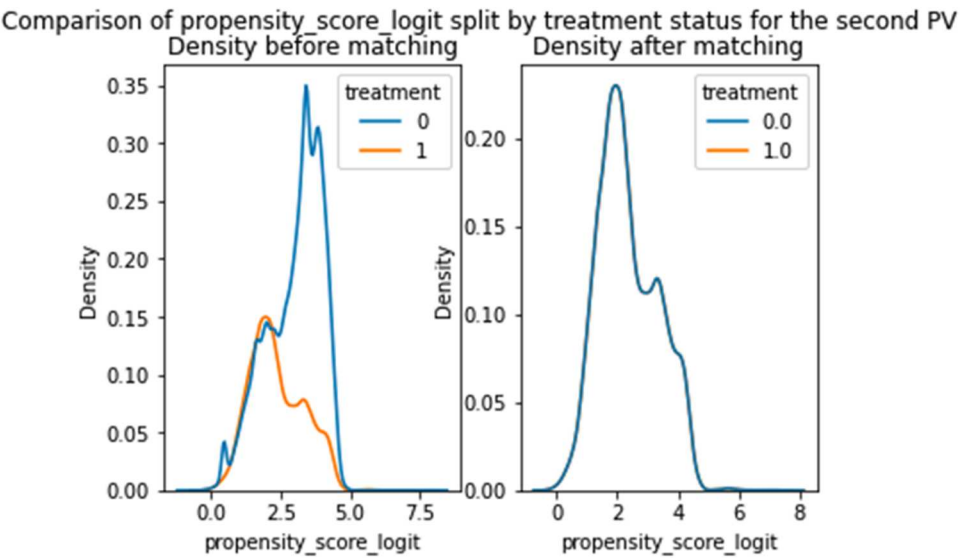


Figure A11. Propensity score logit before and after matching with the second PV.

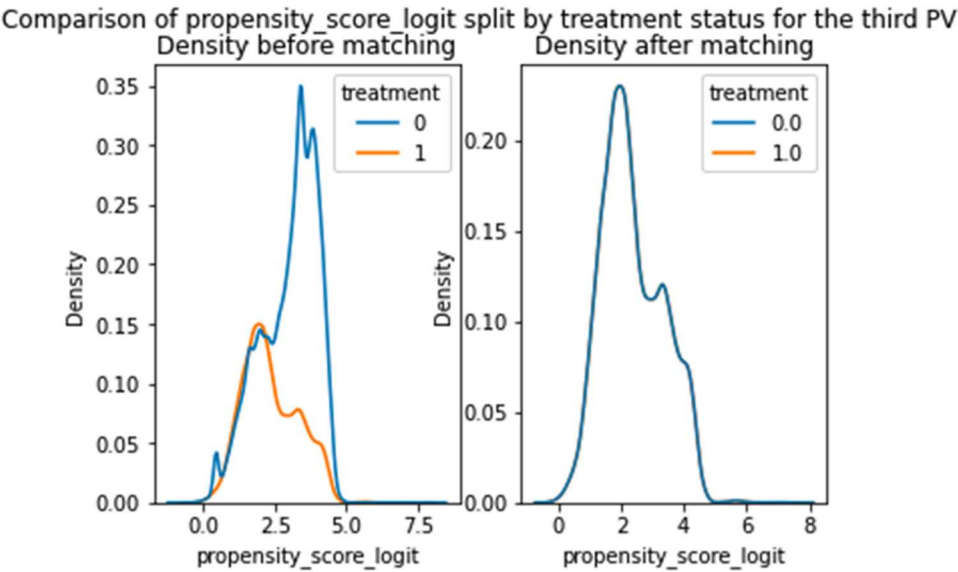


Figure A12. Propensity score logit before and after matching with the third PV.

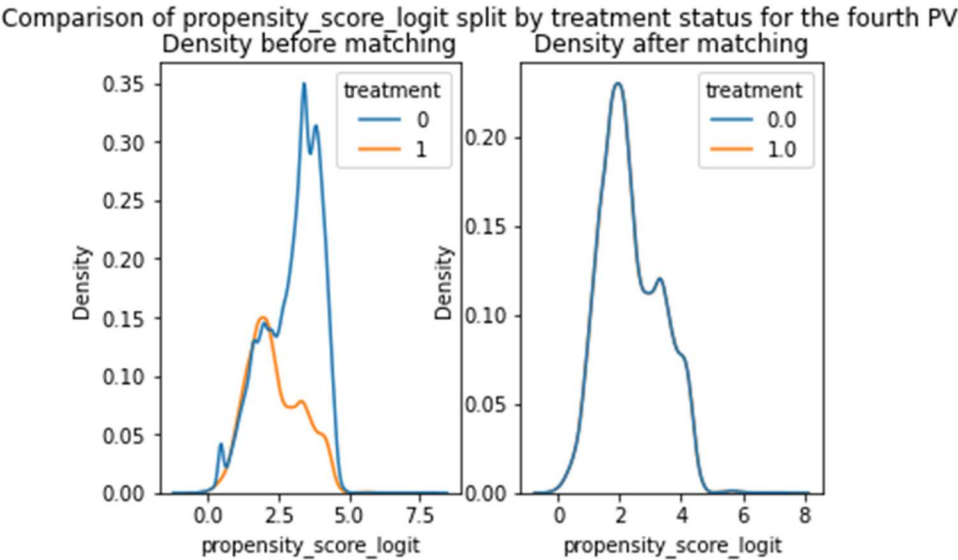


Figure A13. Propensity score logit before and after matching with the fourth PV.

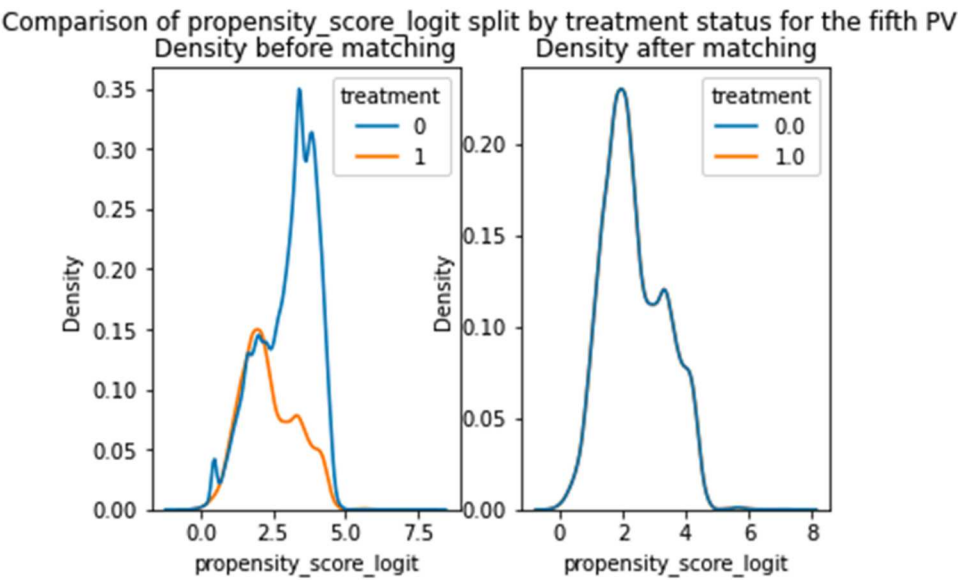


Figure A14. Propensity score logit before and after matching with the fifth PV.

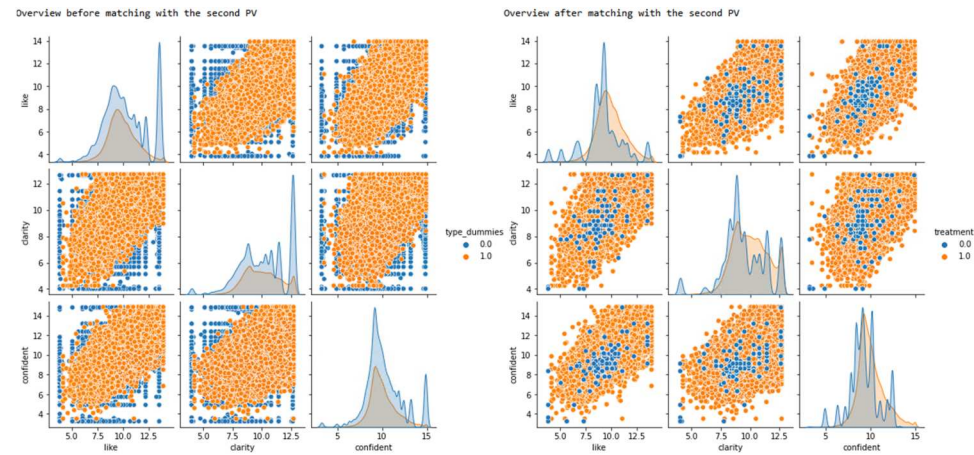


Figure A15. Overview of three covariates before and after matching with the second PV.

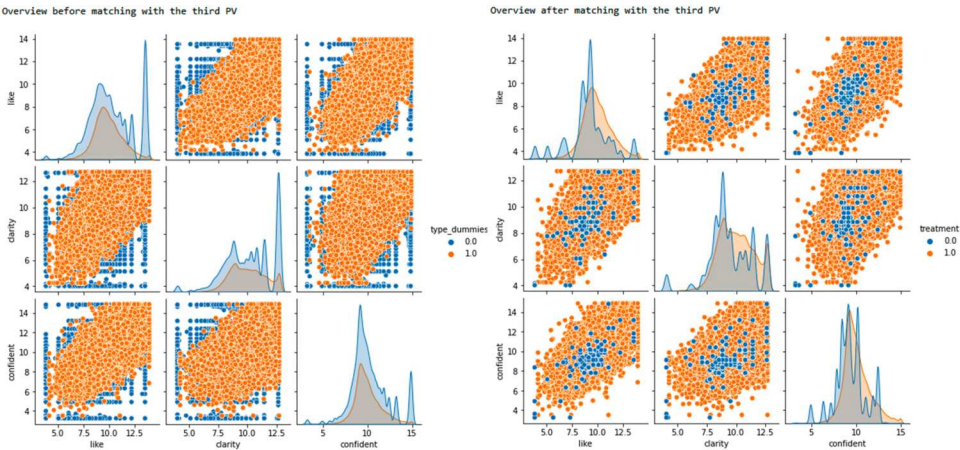


Figure A16. Overview of three covariates before and after matching with the third PV.

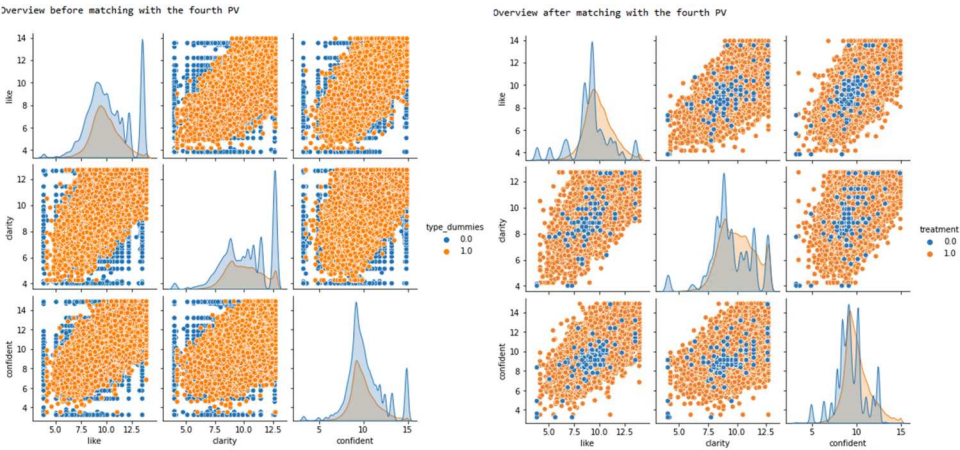


Figure A17. Overview of three covariates before and after matching with the fourth PV.

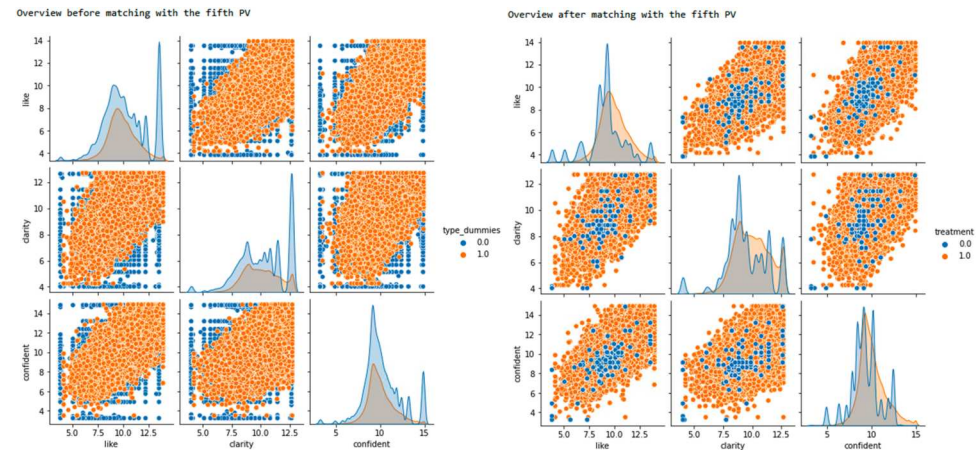


Figure A18. Overview of three covariates before and after matching with the fifth PV.

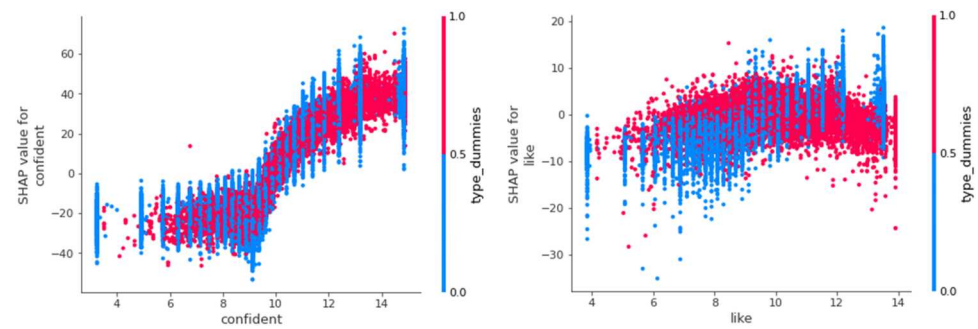


Figure A19. Interactions between student attitudes and curriculum type for the second PV.

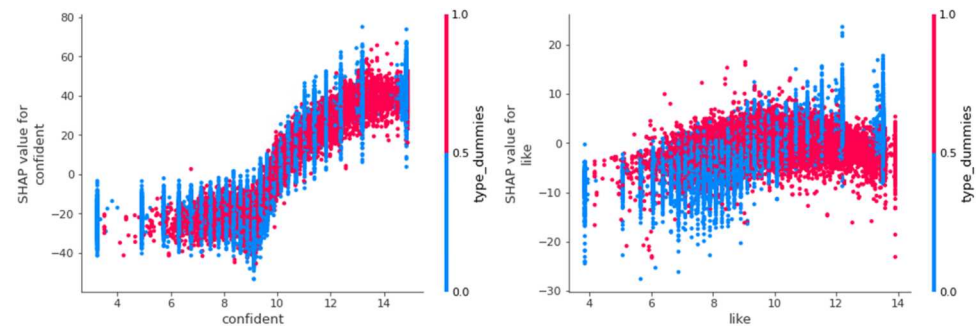


Figure A20. Interactions between student attitudes and curriculum type for the third PV.

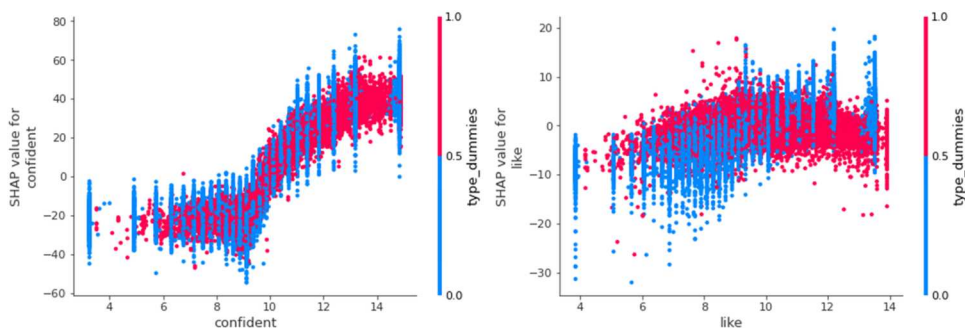


Figure A21. Interactions between student attitudes and curriculum type for the fourth PV.

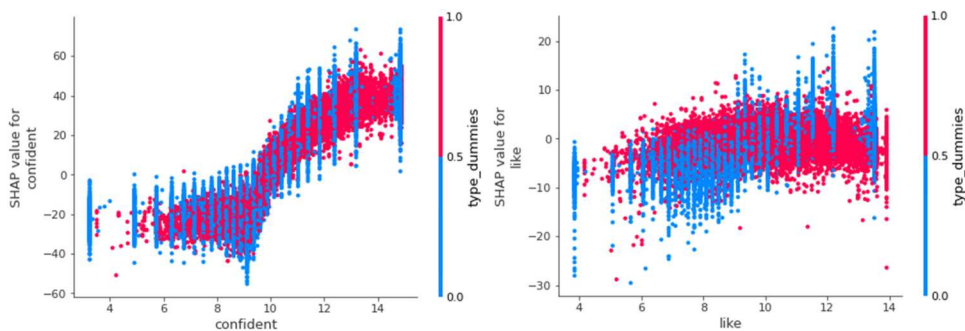


Figure A22. Interactions between student attitudes and curriculum type for the fifth PV.

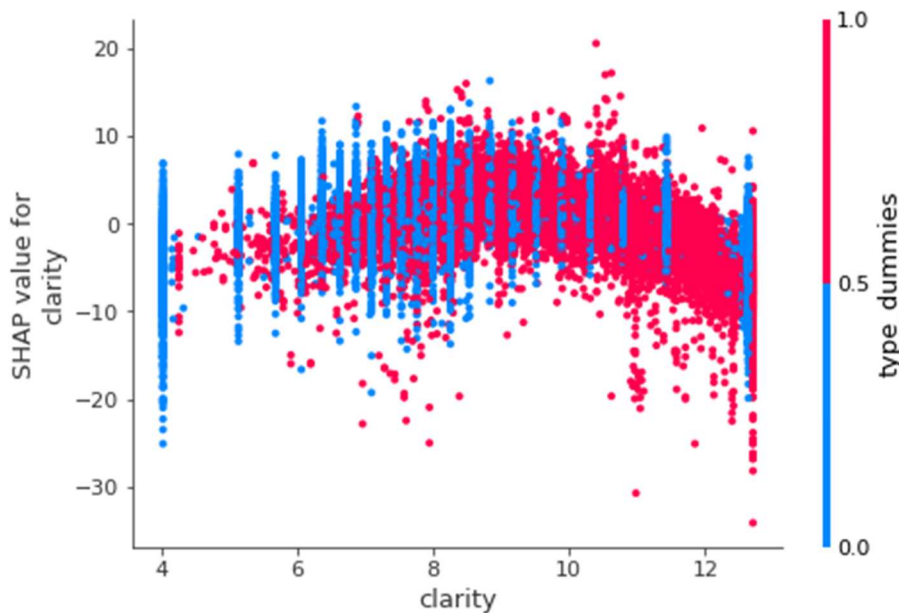


Figure A23. Interaction between instructional clarity and curriculum type for the second PV.

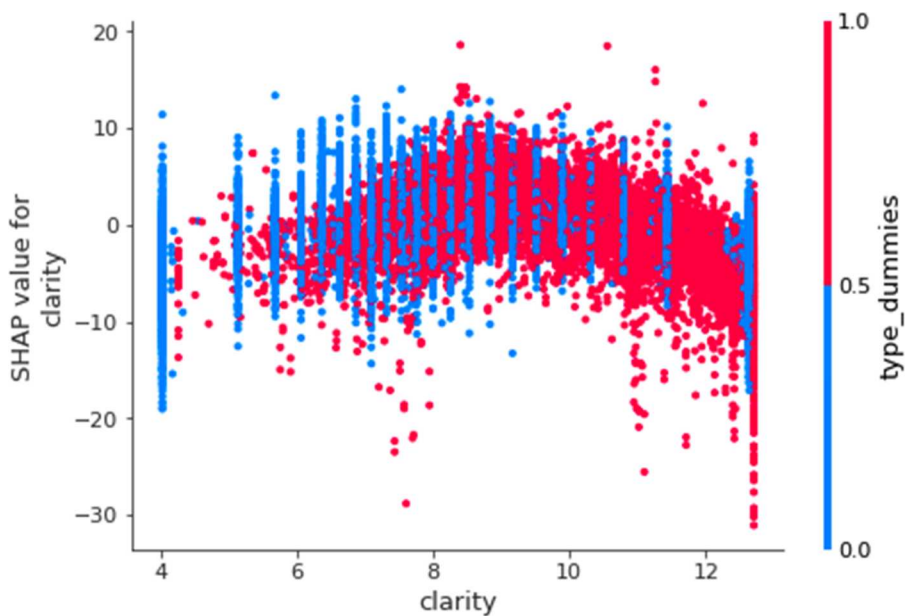


Figure A24. Interaction between instructional clarity and curriculum type for the third PV.

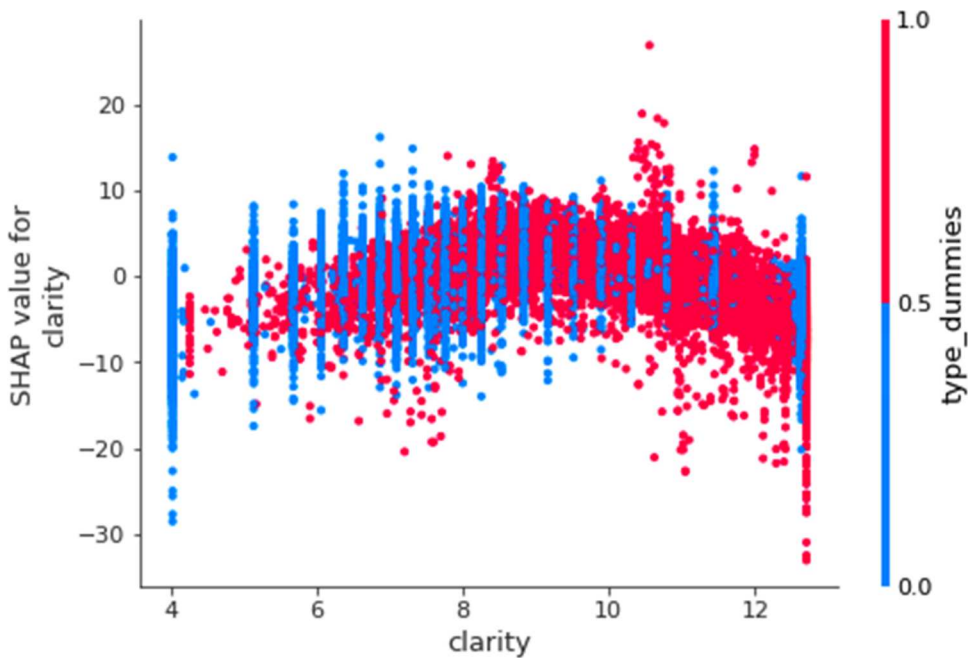


Figure A25. Interaction between instructional clarity and curriculum type for the fourth PV.

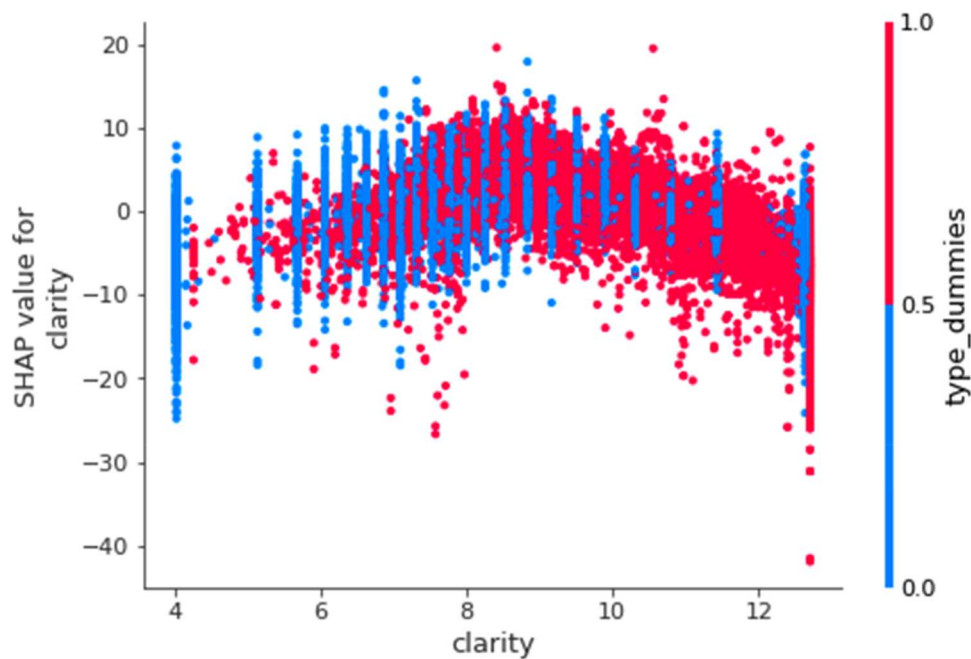


Figure A26. Interaction between instructional clarity and curriculum type for the fifth PV.

Hyperparameter tuning

We employed 10-fold cross-validation to train hyperparameters for each algorithm on the training set, as shown in Table A4. After determining the best model, we tested this model on the test set. After calculating the propensity score for each student, the KNN algorithm was used to match two curricula groups.

Table A1. Abbreviations and the corresponding variable.

Abbreviation	Variable	Abbreviation	Variable
weight	Science teacher weight	like	Students like to learn science
confident	Students feel confident about science	clarity	Instructional clarity
experiments_freq	Science experiment frequency	experiments_like	Students like science experiments
instru_time_fromteacher	Instructional time reported by teachers	instru_time_fromprincipal	Instructional time reported by principals
more_time_prepare	Whether needs more time to prepare instruction	more_time_assist	Whether needs more time to assist instruction
daily_lives	Frequency of linking instruction to daily lives	explain	Frequency of asking students to explain their answers
challenging_exercises	Frequency of asking students to complete challenging exercises	class_discussion	Frequency of encouraging classroom discussions
prior_knowledge	Frequency of linking new content to prior knowledge	problem_solving	Frequency of asking students to find their own problem-solving procedures
express_ideas	Frequency of encouraging students to express their ideas	emphasis_investigation	Teachers emphasise science investigation
homework_freq		ses	Social-Economic background

(Continued)

Table A1. Continued.

Abbreviation	Variable	Abbreviation	Variable
teacher_correct	Homework frequency assigned by teachers	student_correct	Frequency of having students correct their own homework
discuss	Frequency of correcting assignment	monitor	Frequency of monitoring whether the homework was completed
use_for_grades	Frequency of discussing the homework in class	sci_pv	Plausible values of average science achievement
	Frequency of using the homework to contribute students' grades		

Table A2. Decision process for student 31 in the training set.

Student 31			
Prediction: 530.26	Actual value: 649.93	Bias (training set mean): 496.86	
Feature contributions	ses: -19.01	prepared instruction time reported by teachers: 14.34	country: 11.85
	experiments_freq: 7.75	prepared instruction time reported by principals: -6.1	clarity: 5.98
	confident: 5.28	emphasis_investigation: 4.02	like: 3.79
	homework_frequency: -3.72	use_for_grades: -3.38	teacher_correct: 2.93
	gender: 2.26	class_discussion: 2.09	monitor: 1.84
	explain: 1.61	experiments_like: 1.57	problem_solving: -1.47
	curriculum_type: -1.45	prior_knowledge: 1.41	homework_time: 1.12
	express_ideas: 0.90	more_time_prepare: 0.60	daily_lives: -0.57
	challenging_exercies: -0.44	discuss: 0.26	student_correct: -0.15
	more_time_assist: 0.11		

Table A3. Decision process for student 3335 in the training set.

Student 3335			
Prediction 557.28	Actual Value: 614.37	Bias (training set mean): 496.86	
Feature contributions	ses: 26.67	prepared instruction time reported by principals: 13.42	experiments_freq: 10.52
	country: 9.86	prepared instruction time reported by teachers: -7.45	class_discussion: -5.95
	emphasis_investigation: 5.51	prior_knowledge: 4.54	clarity: -4.13
	curriculum_type: -3.23	express_ideas: 2.81	homework_freq: -2.26
	like: 2.13	more_time_assist: 2.12	confident: 1.79
	student_correct: 1.44	explain: 1.07	challenging_exercies: 1.03
	teacher_correct: -0.97	daily_lives: 0.95	homework_time: 0.87
	use_for_grades: -0.78	more_time_prepare: 0.70	discuss: -0.60
	problem_solving: 0.52	monitor: -0.20	gender: 0.12
	experiments_like: -0.06		

Table A4. Hyperparameters of propensity score calculation.

Algorithm	Parameters	Range	Optimized value
KNN	n_neighbors	1, 5	1
Random Forest	n_estimators	10, 50, 100	100
Logistic Regression	C	0.1, 10	10

Table A5. Hyperparameters of science achievements prediction.

Algorithm	Parameters	Range	Best Value 1	Best Value 2	Best Value 3	Best Value 4	Best Value 5
Lasso	alpha	linspace (0.1, 50, 50)	0.1	0.1	0.1	0.1	0.1
Decision tree	max_depth	3, 6, 9	9	9	9	9	9
	max_features	randint (1, 25)	23	22	22	22	23
	min_samples_leaf	randint (1, 25)	11	21	21	21	11
	criterion	['squared_error,' 'frideman_mse']	friedman_mse	friedman_mse	friedman_mse	friedman_mse	friedman_mse
Random forest	n_estimators	100, 350, 500	500	500	500	500	500
	min_samples_leaf	3, 4, 5	3	3	3	3	3
	min_samples_split	8, 10, 12	8	8	8	8	8
	max_features	['log2,' 'auto,' 'sqrt']	auto	auto	auto	auto	auto
	max_depth	4, 8, 12, 16, 20	20	20	20	20	20
Neural networks	batch_size	10, 20, 30	20	10	10	20	10
	epochs	10, 20, 100	100	20	20	100	100
	Optimizer_trial	['adam,' 'rmsprop']	rmsprop	rmsprop	rmsprop	rmsprop	rmsprop

Note: The best values from 1 to 5 represent the parameters derived from the fittings with five respective PV.